



Ministry of Digital Economy and Entrepreneurship
MoDEE

**Request for Proposal (RFP) Document for Implementation
of Integrated Tax Administration Solution for Income tax
and Tax Directorate**

**Volume 2B: Integrated Tax Administration Solution -
Terms of Reference**

Deadline: 00/00/2025

RFP No.: XXXXX

Abbreviations

No.	Term	Definition
1	Bidder(s)/Participant(s)	Shortlisted firms submitting their proposal as a single bidder or as Consortium with multiple participants in response to this RfP document
2	BRD	Business Requirement Document
3	EDI	Electronic Data Interchange
4	ERP	Enterprise Resource Planning
5	FRD	Functional Requirement Document
6	IaaS	Infrastructure as a Service
7	ISTD	Income & Sales Tax Department
8	ITAS	Income Tax Administration System
9	MoDEE	Ministry of Digital Economy and Entrepreneurship
10	PaaS	Platform as a Service
11	RTM	Requirement Traceability Matrix
12	SaaS	Software as a Service
13	SLA	Service Level Agreement
14	SRS	System Requirement Specifications
15	UPC	Universal Product Code
16	OECD	Organisation for Economic Co-operation and Development
17	SANAD	Mobile App for Digital Government Citizen Services
18	IDM	Identity Management

About this Volume 2B: Integrated Tax Administration Solution ToR

This document provides information regarding the scope of work including business requirements, software and services to be covered and corresponding related documentation, scope of work for the Winning bidder, Functional requirements & Technical requirements, Manpower Requirements, Payment Terms and Service level Agreements, which must be adhered to by the Winning Bidder while delivering the for implementation of Integrated Tax Administration Solution at the Income & Sales Tax Department.

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1. Introduction

Over the last two years, the Income & Sales Tax Department (ISTD) and the Ministry of Finance (MoF) have made significant progress in the modernization of legislation enabling focused tax reforms and addressing distortions and imbalances in a way that can improve tax justice, raise audit efficiency, combat tax evasion and address tax avoidance instead of imposing new or increasing taxes.

With the successful reforms delivered in the areas of auditing, control and follow-up procedures combined with the result of electronic linking of information with all institutions has increased revenue from both direct and indirect taxes, the Government of Jordan is committed to an economic reforms agenda which, amongst other things, targets the improvement in the performance and efficiency of the revenue system and strengthening the public budget management processes. At the same time, the Government is keen not to raise taxes from the current levels but rather achieve public revenue targets by supporting the growth of the private sector with legislative stability and transparency.

These changes have been introduced with an overarching view to strengthen the culture of partnership between ISTD and taxpayers, improving the quality and standards of its services to that of matured jurisdictions, optimize its operations and seek to leverage the latest technologies to address issues around tax evasion and avoidance. Taking this further, a decision has been made to implement an Integrated Tax Administration System (ITAS) as an 'initiative' as part of its tax administration strategy.

The Income Tax Department and the General Sales tax Department were merged into one Department during 2004 under the name of Income and Sales Tax Department ("ISTD"). The ISTD governs, administers, and manages different types of taxes. Entities undertaking operations in Jordan are required to comply with all tax and social security registration, reporting and filing requirements, including but not limited to, corporate income tax ("CIT"), personal employee income tax ("PIT"), withholding tax ("WHT"), national contribution tax, General and Special Sales Tax ("GST" & "SST"), social security, stamp duty and customs duty.

Table 1: Key Data Points - ITAS

Total number of registered (Income and Sales Tax) Taxpayers		1,826,000
Total number of registered (Income and Sales Tax) taxpayers - Active		540,000
Registered "Individuals"		806,000
Registered "Individuals" - Active		195,000
Registered "Employees"		783,000
Registered "Employees" - Active		265,000
Registered "Companies"		140,000
Registered "Companies" - Active		40,000
Registered "Partnership Companies" - Active		40,000

Table 2: ISTD users

ISTD users	Count
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Number of ISTD user logins	1,600
Number of active ISTD users	1,419

Note: The bidder needs to factor an annual growth of 10% in above numbers for the purpose of application, infrastructure sizing and any other related items, as deemed appropriate by the bidder for optimal operations and management of SLA for 10 years of maintenance period. Refer Annexure E – ITAS Technology Stack

While ITAS will help ISTD in the administration of all existing taxes it currently handles, it is also expected that ITAS will be tax agnostic and have the capability to administer additional taxes, if decided to be introduced by the Government of Jordan in the future. The bidder is expected to provide their ITAS business process maps complying with the international best practice for Tax Administration which will be adopted by ISTD with necessary customizations as required.

1.1. ITAS Conceptual Model

ITAS will address the tax administration requirements of ISTD as well as the tax compliance requirements of taxpayers throughout the ISTD's value chain. A conceptual model of ITAS is shown below.

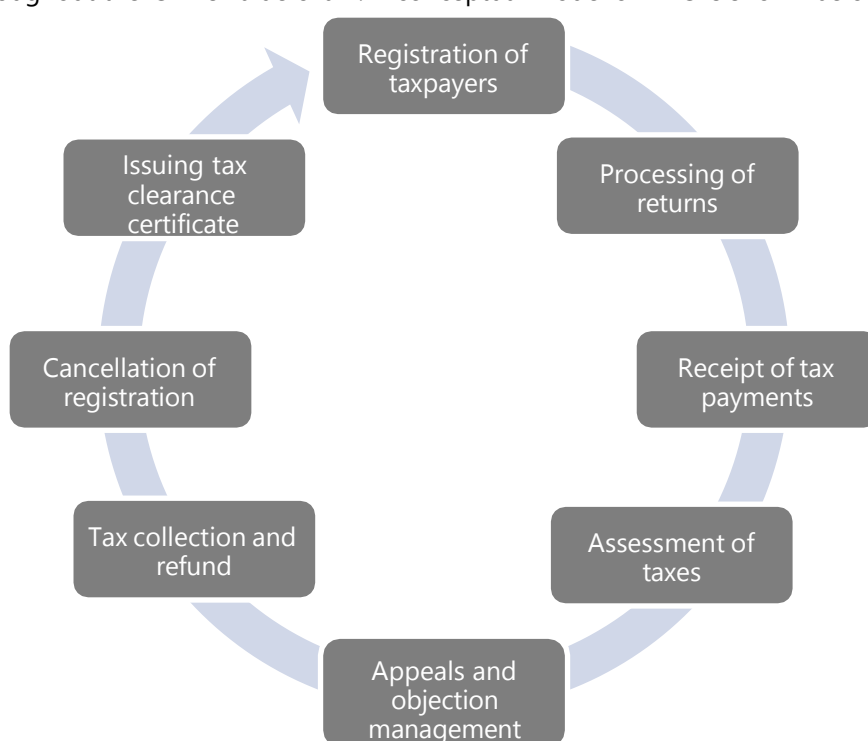


Figure 1: Tax administration lifecycle

ITAS conceptual model identifies different integral component of ITAS such as the users, service delivery channels, types of services, and functionalities to be supported. This section briefly discusses each of these components to provide an overview of the conceptual model of ITAS.

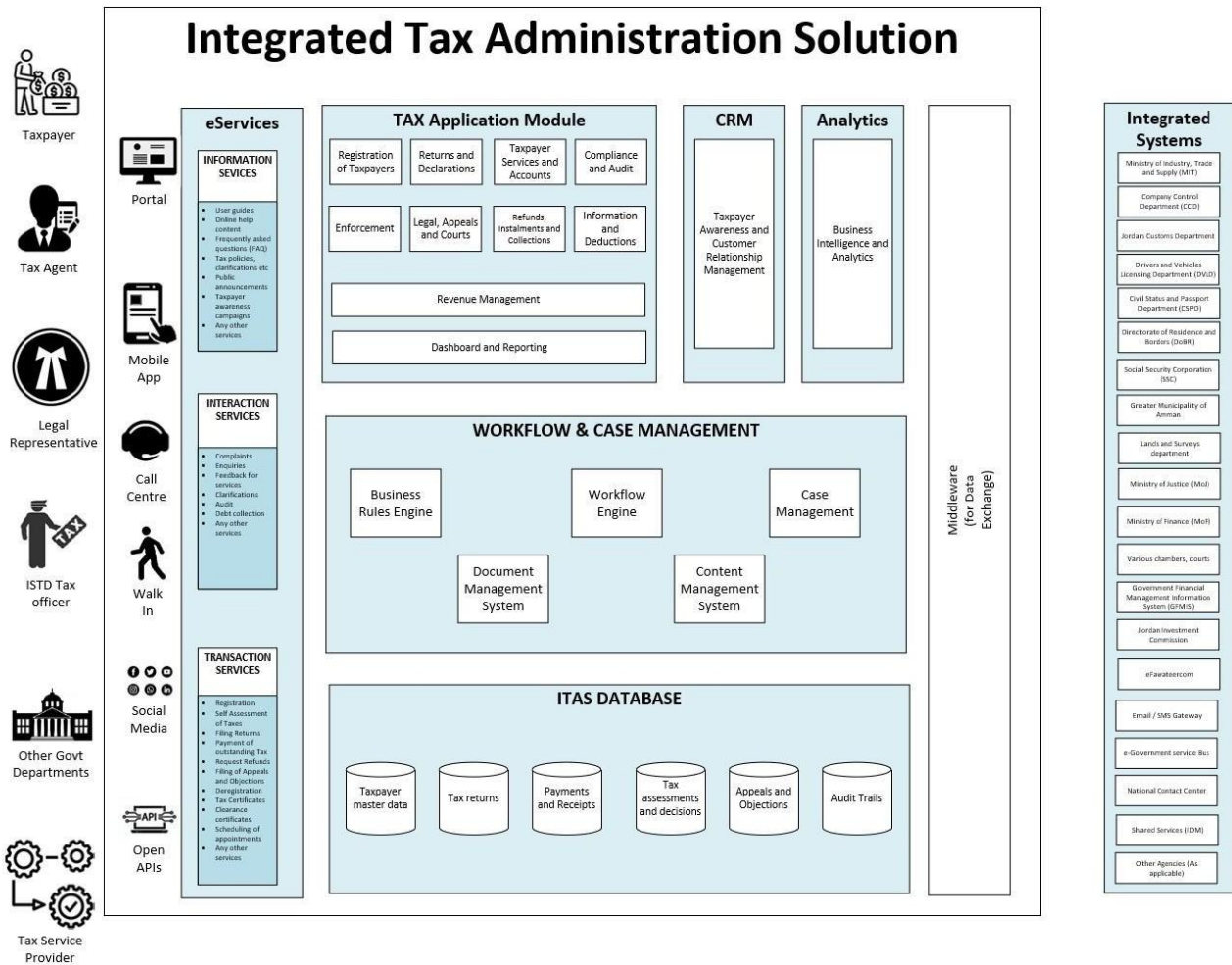


Figure 2: ITAS Conceptual model

1.1.1. ITAS Users

The below table provides an overview of the various categories of users who interact with ITAS. (Note: the bidder shall determine the final list as part of the business requirements gathering phase of the project).

Table 3: ITAS Users

Category	How they interact with ISTD?
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Taxpayer	Taxpayers are the key users and beneficiaries of ISTD. The system scope and design shall focus on the needs of taxpayers in improving the quality and effectiveness of service delivery. Taxpayers will have the facility to transact with ISTD through multiple service delivery channels. The system will provide significant support in delivering self-services to taxpayers, thereby requiring minimal physical interaction between taxpayers and ISTD. The Taxpayers registered under the corporate legal framework in Jordan are classified into following: Natural Person ▶ Personal (Employee) - A natural person who receives employment salaries and benefits and has other sources of income (such as income from investment, trade, sole proprietorship)
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Category	How they interact with ISTD?
	▶ Individual (User) – A natural person who receives employment salaries and benefits only Legal Person ▶ Partnership ▶ General Partnership ▶ Limited Partnership ▶ Limited Partnership in Shares ▶ Company ▶ Limited Liability Company ▶ Private Shareholding Company ▶ Public Shareholding Company
Tax agent	Tax agents can make use of ITAS, to transact with the ISTD on behalf of their Taxpayers. Taxpayers can nominate, via the e-Services portal, registered tax agents to perform transactions and represent them with ISTD. Such nominated tax agents, on behalf of their clients (taxpayers), can perform transactions such as filing returns, filing appeals, and providing clarifications to the queries of ISTD.
Legal representative	A legal representative is designated for a taxpayer via a power of attorney or court order appointment to act as that taxpayer's representative. Once so designated, the legal representative acts on behalf of the taxpayer in most situations, from tax return submission through compliance, payment, filing appeals, and providing clarifications to the queries of ISTD.

Other departments, banks	Government agencies,	Various Jordan government departments work with the ISTD to collaborate on tax administration depending on their specific responsibility. A high-level list of such personas is given below. (Note: This is not an exhaustive list* and the bidder needs to finalize as part of the Business Requirements documentation during the engagement) ▶ Greater Amman Municipality ▶ Amman Chamber of Commerce ▶ Amman Chamber of Industry ▶ Minister of Digital Economy and Entrepreneurship ▶ Department of Lands and Survey ▶ Banks ▶ Driver and Vehicle Licensing Department ▶ Petra Development & Tourism Region Authority ▶ General Supplies Department
Category		How they interact with ISTD?
ISTD tax officer		▶ Jordan Standards and Metrology Organizations and Ministry of Energy *Refer ITAS Annexure C – Interfacing Systems for a list of existing integrations (As-Is) and interfacing systems (To-Be) ISTD staff must utilize ITAS to process registrations, returns, manage compliance programs, and respond quickly to requests for support from taxpayers, tax agents, legal representative, other government departments.

1.1.2. Service Delivery Channels

One of the key objectives behind ITAS is to improve the operational convenience for taxpayers by delivering services through different channels and enabling them to address their compliance requirements. An indicative list of the service delivery channels that should be considered for ITAS implementation is shown below:

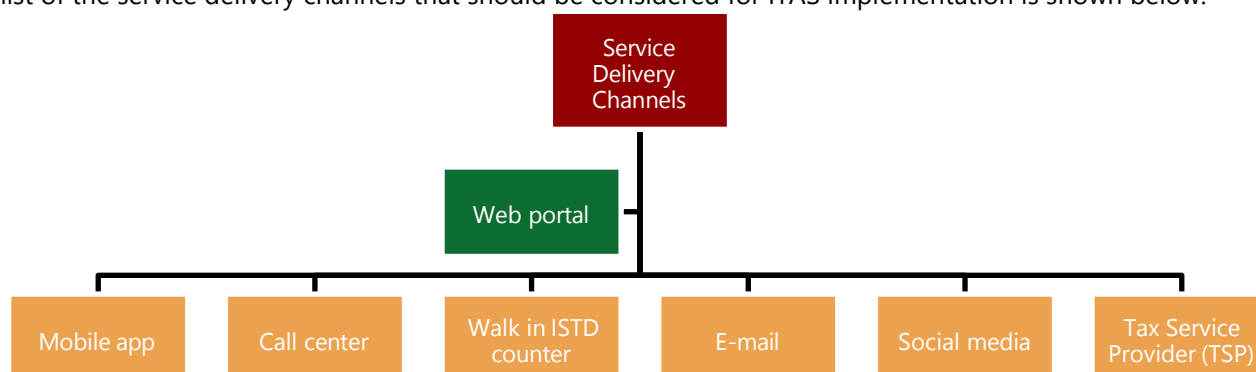


Figure 3: ITAS Service delivery channels

Table 4: Service Delivery Channels

Delivery Channel	Description
Web portal	The ITAS web portal is the key service delivery channel to support the ISTD in the conversion of its services into self-services, which can be performed by taxpayers without visiting the offices of the ISTD. The portal will be an interactive channel and play a crucial role in providing information, interaction, and transaction services to taxpayers and other ITAS user categories explained above.
Mobile app	ITAS will also provide mobile app for the above listed user categories to access the eServices via mobile devices and to perform a limited number of information, interaction, and transaction services through mobile devices. In addition to the ITAS mobile app some of the services will also be available in SANAD App.
Taxpayer service call center	ISTD call center will enable taxpayers to seek necessary clarifications on the taxpayer services and get support on ITAS. In case the taxpayers call the National E-Government Contact Center of Jordan (NCC) then NCC is expected to re-route such calls received from the taxpayers to the ISTD call center.
Walk-in counters (Face to Face)	Taxpayers can visit the walk-in ISTD counters for application submission, tax-related inquiries/complaints, payment (including cash/cheques) and document delivery if needed and all needed processes that require the taxpayer presence. ITAS shall enable ISTD staff at the department counters to use the ITAS for delivering the taxpayer services.
E-mail support	E-mail support will enable taxpayers to seek clarification or guidance on taxpayer services and shall facilitate the reporting of errors such as log-in issues related to the system's use.
Social media support	The social media (e.g.: WhatsApp) delivery channel under ITAS will enable taxpayers to seek necessary clarifications on the taxpayer services of the ISTD. In addition to WhatsApp, additional social media handles support may need to be enabled based on business requirements gathering discussions with ISTD.
Tax Service Provider (TSP)	Empaneled group of service providers made up of multiple Tax Service Provider (TSP) who will use the services of ITAS (using APIs) and provide enhanced services to the end users.

1.1.3. ITAS – Information Flow

The diagram below provides a high-level information flow of the business capabilities described above

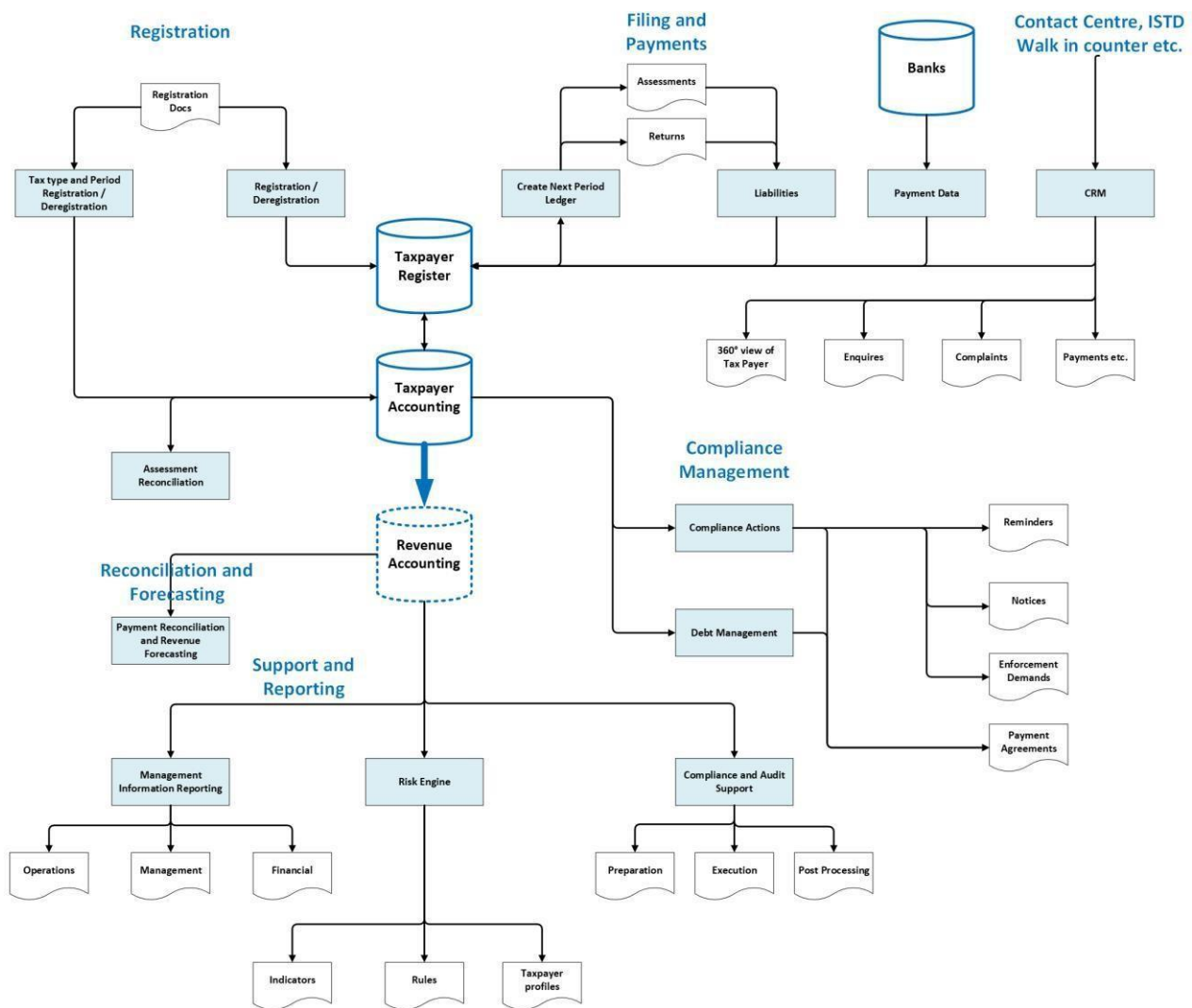


Figure 4: ITAS Information Flow

1.2. ITAS Services and Functions

ISTD provides a broad range of services and functions related to their tasks. It can be everything from pure dispersion of information, via different types of guidance services, which often include a dialogue between ISTD staff and users, to different types of transactional services. These types of services are relevant to channel strategy discussions since some channels may be more suitable than others for providing different types of services, and the users, citizens, and organizations, may have different channel preferences, based on who they are and which service they require. The transaction services are the “core business” of ISTD, and the information and interaction services are services supporting the transaction services.

ITAS must enable ISTD to approach the services in a structured way, and which is of relevance to channel use and users, the following categorization may be considered:

- ▶ Information services
- ▶ Interaction services
- ▶ Transaction services

1.2.1. Information Services

Information services refer to delivering information to taxpayers related to the services and functions of the ISTD, and the compliance requirements of the taxpayers, in a convenient and easily accessible manner. The information services will be delivered to taxpayers through service delivery channels such as a portal, mobile portal, call center, and walk-in ISTD counters. The following is an illustrative list of information services proposed to be delivered through ITAS in both English and Arabic. Note: The detailed list needs to be derived as part of the requirement gathering workshops in the project.

- ▶ User guides
- ▶ Online help content
- ▶ Frequently asked questions (FAQ)
- ▶ Tax policies, clarifications etc.
- ▶ Public announcements
- ▶ Taxpayer awareness campaigns
- ▶ Taxpayer correspondences
- ▶ Any other services (identified during requirement gathering)

1.2.2. Interaction Services

Interaction services are two-way information exchange between taxpayers and the ISTD, which does not result in any change in the taxpayer's account status. The following is an illustrative list of interaction services proposed to be delivered through ITAS in both English and Arabic. Note: The detailed list needs to be derived as part of the requirement gathering workshops in the project.

- ▶ Complaints
- ▶ Enquiries
- ▶ Feedback for services
- ▶ Clarifications
- ▶ Audit
- ▶ Debt collection
- ▶ Any other services (identified during requirement gathering)

1.2.3. Transaction Services

Transaction services refer to the activities or services that result in a change in taxpayer's account status or account information. The following is an illustrative list of transaction services proposed to be delivered through ITAS in both English and Arabic. Note: The detailed list needs to be derived as part of the requirement gathering workshops in the project.

- ▶ Registration
- ▶ Self-Assessment of Taxes
- ▶ Filing Returns
- ▶ Payment of outstanding Tax
- ▶ Request Refunds
- ▶ Filing of Appeals and Objections
- ▶ Deregistration
- ▶ Tax Certificates
- ▶ Clearance certificates
- ▶ Scheduling of appointments
- ▶ Any other services (identified during requirement gathering)

1.2.4. Tax Experience Management

The various business functions to enrich the Taxpayer & Tax officer User Experience as listed as below

Personalization

The user must be allowed to access their personalized Home page through their user credentials after being validated by the portal. The respective user should be able to see and perform actions in the respective channels (front end or back end) based on the roles and access assigned to them.

Content and Document Management

The ITAS should enable the ISTD to automate the Taxpayer business functions to create a fully digital platform by reducing the manual paper-based Tax processing. The Content and Document management system should be capable of storing and retrieving documents related to Public Announcements, User Manuals, Taxpayer Registration, Certificates, Returns filed, Refunds issued, Payments made, Inspections completed, Objections raised, Decisions made, Clarifications provided through a well-defined folder structure enabling proper version management. Various users should have access to content based on their user-based roles. This Document management system should also migrate all the existing structured and unstructured data in both physical and digital form to fit into the newly proposed folder structure to serve as a single repository for all the data. This should also serve as a single source to all front-end applications like ITAS portal, ISTD Jordan website, Tax Service Providers etc.

1.2.5. Customer Relationship Management (CRM)

The Customer Relationship Management is a repository of all contacts with taxpayer and should be integrated

with Jordan's e-Government National e-Government Contact Centre. When a taxpayer makes contact via phone, the contact center staff must get an instant on-screen retrieval of all previous contacts with the taxpayer based on their tax identification number. The contact registered in ITAS can be correspondence, email, phone, walk-in visit, or any kind of self-service, including SMS text messages. (In addition, self-service transactions by the taxpayer that failed are also recorded). This provides a detailed contact history at customer level and makes it possible to perform analysis of contact channel trends etc. The contact should categorize according to predefined categories such as degree of complexity, whether urgent or not, region, status, and some other categories. CRM should also support the work of the ISTD in other ways such as service request management (including inquiries and complaints) and public information campaigns about initiatives of the ISTD.

1.2.6. Workflow and Case Management

Workflow management is the integral component of ITAS that not only facilitates workflow but also supports other functions such as document and content management. Some of the features of the workflow management tool are given below.

- ▶ Workflow engine supports work management at the operational level, Self-Assessment of Taxes including creating activities, assigning work, reporting, escalation, and administration of workflow.
- ▶ Business Rules Engine is a centralized business rule repository for storage and management of business rules and data validation rules.
- ▶ Document management provides the ability to store and retrieve electronic versions of documents and manage the retention and destruction of electronic records in accordance with legislative requirements.
- ▶ Content management is the enterprise functionality for initiating, approving, storing, maintaining, and publishing approved content for informational, transactional, interpretative, and correspondence products, including marketing and education materials.
- ▶ Case management manages all types of workflow for an individual case, e.g., a taxpayer audit, a taxpayer debt, a disputed assessment, an outstanding return, and responding to taxpayer correspondence.
- ▶ Generate determined management information for all case types, such as cases created, cases closed, cases still outstanding, elapsed time for actions, and maintain an audit trail of any accesses or adjustments made.

1.2.7. Interfacing with external systems

The ITAS will be designed as a faceless system with API-driven architecture at its core. This will enable the access of all system features via any user interface (internal or third-party applications) that work on top of these APIs. The ITAS portal and mobile app will be applications on top of these APIs. ITAS shall integrate with the various Ministry and third-party systems/databases as shown in the below diagram.

2. ITAS Functional Requirements

The below diagram shows the Business Capability Model for the ISTD. It is expected that the ITAS will support these business capabilities, associated processes, requirements, Specifications, and functionalities.

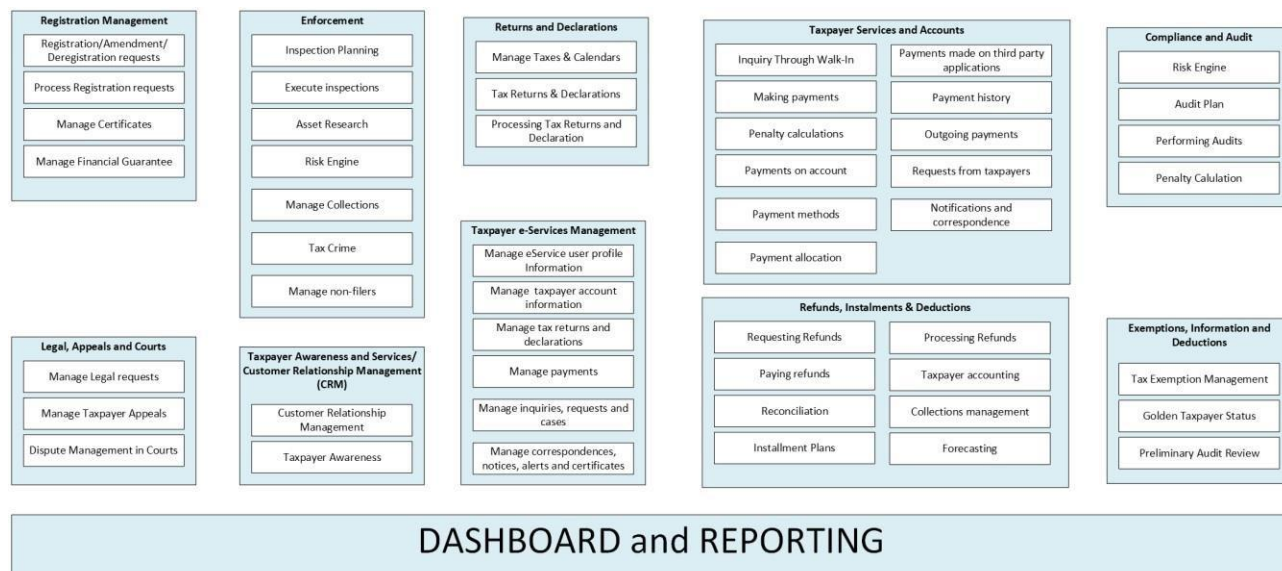


Figure 5: ITAS Functional Modules

The sections below provide an overview of each functional module in ITAS. Refer to the detailed requirements for each module in the ITAS Annexure A - ITAS Functional Requirements. Bidder is expected to update the 'requirement compliance' column of this annexure as part of response submission.

2.1. Taxpayer e-Services Management

The e-Services are an online interface (web, mobile app) to the end-users for enabling self-services in the ITAS. Key features are:

- Self-services to minimize the effort of taxpayers in complying with tax laws and to reduce the administrative burden for ISTD tax officials in processing transactions.
- Minimize the cost of tax compliance for taxpayers and the cost of tax administration for the ISTD. Enable an integrated view on taxpayer information and tax liabilities to the taxpayer.
- Enable more efficient communications between taxpayers and ISTD Tax Officers.

Table 5: Taxpayer e-Services Management

Workstream (Level 0)	Functional (Level 1)	Area Functional Area Definition
Taxpayer e-Services Management	Manage eService user profile Information	The ability for a user to, login using SANAD credentials , setup profile preferences, setup access levels, link/delink profile to taxpayer account etc.

Manage taxpayer account information	The ability for the taxpayer to get a 360 degree summary view of their information such as taxpayer information (e.g. name, address, contact information, etc.), tax status (e.g. outstanding balance, open enforcement actions, open audits, etc.), tax information (e.g. returns filed, payments made, refunds issued), history (e.g., correspondence issued, customer services notes, etc.) etc.; complete and submit registration application for different tax types, amendment application for change in circumstances, deregistration application to close taxpayer account
Manage tax returns and declarations	The ability for the taxpayer to complete, submit, amend, and manage tax returns, declarations, and statements
Manage payments	The ability for the taxpayer to make payments for tax returns and obligations, setup preferred payment methods, setup payment allocation rules, view payment history etc.
Manage inquiries, requests, and cases	The ability for the taxpayer to complete and submit general inquiries and requests such as refunds, objections, appeal, tax clearance, payment plan etc., track status and view history
Manage correspondences, notices, alerts, and certificates	The ability for the taxpayer to view and download correspondences, notices, alerts and certificates, view mass communications sent by ISTD

2.2. Registrations Management

The registration of taxpayers will facilitate online registrations for applicable tax types, the allocation of Tax Identification Numbers (TIN), among other functions. Taxpayer registration will ensure the unique identification of individual taxpayers to enable the efficient conduct of all downstream administration processes and provide an integrated view of taxpayers, including tax liabilities and payments across various tax categories. Key features are:

- ▶ Issue and maintain a national tax identification number for taxpayer registrations.
- ▶ Calculate and maintain a check digit and internal referencing to ensure the validity of these identification numbers.
- ▶ Capture and store specific registration details for different tax types in a 'whole of client' view.
- ▶ Integrate the registration system with all other core business functions, e.g., returns and payment processing, so that registration details are stored and maintained once, but reused by other systems.
- ▶ Deregister taxpayers and archive information in such a way that it can be restored.

- ▶ Allocate one identification number for a legal entity but enable branches of an entity to be linked.
- ▶ Allow for linking details where taxpayers are directors of companies and or are partners in partnerships.
- ▶ Generate registration management information, e.g., registrations by entity type, by office, by region, by sector, by industry, etc. and an audit trail of any accesses and adjustments made

Table 6: Registrations Management

Workstream (Level 0)	Functional (Level 1)	Area	Functional Area Definition
Registration Management	Registration/Amendment/Deregistration requests		The ability for the taxpayer to complete, submit, amend, and manage registration/amendment/deregistration applications
	Process Registration requests		The ability for the system to apply business rules and route the workflows to the ISTD tax officers to review and approve/reject the taxpayer registration applications and register the taxpayers
	Manage Certificates		The ability for the system to manage the certificates including maintaining certificate templates and versioning
	Manage Financial Guarantee		The ability for the system to manage the financial guarantees submitted by the taxpayers.

2.3. Returns and Declarations

Taxpayers in using online services to compute the accurate amount to be paid to ISTD based on liable revenues. Electronic filing of returns will facilitate taxpayers to prepare and file the returns, for all the applicable taxes, electronically and to submit returns online. Taxpayers can reuse the information related to the taxpayer already recorded in ITAS, through various other sources for filing returns, which minimizes the effort of taxpayers and increases the scope of monitoring and compliance for the tax agency. Key features are:

- ▶ Process tax forms (declarations/returns) for all tax types.
- ▶ Support fully the principles of self-assessment for processing declarations/returns, including: (1) check taxpayer identity against the taxpayer registration system; (2) record date of filing; (3) acceptance of the tax liability declared by the taxpayer;(4) perform arithmetical calculation based on the data in the form (declaration/return);(5) compare information submitted against information held within the database for the taxpayer; and (6) storage of tax form (declaration/return) data in the database.
- ▶ Be configured easily to cater for new tax types or changes to existing taxes.
- ▶ Be able to perform a reconciliation of third-party income details with information declared.
- ▶ Receive forms (declarations/returns) electronically.
- ▶ Generate electronic receipt for electronic tax forms (declarations/returns).
- ▶ Process amended tax forms (declarations/returns) or assessments.
- ▶ Issue assessment notices electronically.
- ▶ Raise default assessments where a tax form (declaration/return) has not been lodged where appropriate.
- ▶ Event calendar capability to
- ▶ Determine whether a form (declaration/return) for a particular tax type should be expected from a taxpayer. This includes a 'nil return'. Automatically generate appropriate returns for each Revenue type.
- ▶ Determine the due date for all tax form (declaration/return) types according to legislation and administrative arrangements.
- ▶ Allow an extension to the due date for filing/lodgment of a form (declaration/return) if a taxpayer's request is approved.
- ▶ Automatically generate a request/demand for the form (declaration/return) after a nominated period, after the form (declaration/return) has not been filed.
- ▶ After a determined period, automatically issue a more strongly worded demand if the form (declaration/return) has still not been filed.

- ▶ After a further determined period, if the form (declaration/return) is still outstanding, allocate the case for manual follow up via the case management system.
- ▶ Archive, to an environment that can be restored if required, form (declaration/return) information where the account balance is nil at an appropriate determined time.
- ▶ Produce management information on quantity of forms (declarations/returns) received, and an audit trail of any accesses or adjustments made.
- ▶ Generate management information reporting on how many forms (declarations/returns) were expected, how many were received, how many are still outstanding, the age of the outstanding forms (declarations/returns) for all offices and be able to consolidate this information on a national basis. An audit trail of any accesses or adjustments made is also required.
 - The proposed system should cover transfer pricing (profit shifting) treatment.

Table 7: Returns and Declarations

Workstream (Level 0)	Functional (Level 1)	Area	Functional Area Definition
Returns and Declarations	Manage Taxes & Calendars		The ability for the system to manage different tax types, tax rates, filing calendars, reminders, method of filing and correspondences based on ISTD procedure.
	Tax Returns & Declarations		The ability for the system to enable taxpayers to file different tax returns and declarations applying multiple tax rates and rules
	Processing Tax Returns and Declaration		The ability for the system to perform various checks taking inputs from various systems, following defined rules of tax type and workflows.

2.4. Payments

Electronic payment of taxes will facilitate the recording of the details of payments made at banks or through the ITAS (portal/mobile channels). This also supports an internet payment facility via National Payment gateway (eFAWATEERcom), along with inter banking transfers (e.g., electronic fund transfers) for the purpose of payment of taxes. Payments can also be made via cash/cheques by the Taxpayer physically at the walk-in ISTD counter. Provision for managing payment requirements, including deadlines (or due dates) for payment, who is required to pay and payment methods. Provision in the system to identify payments due, either self-assessed or administratively assessed. Failure by a taxpayer to pay on time results in imposition of interest and penalties and follow-up action by the tax administration, including legal debt recovery action. Key features are:

- ▶ Capture identification number, tax type, payment period, payment type, and payment amount.
- ▶ Update these details into the taxpayer and revenue accounts automatically.
- ▶ Automatically interface with external payment method options, e.g., banks, "e-payment." via eFAWATEERcom
- ▶ Reverse payment data and manage any re-transmissions of payments data including the processing of dishonored payments.
- ▶ Process a payment to a suspense account if taxpayer or account details are not known.
- ▶ Generate management information that enables:
 - ▶ payments reconciliations.
 - ▶ revenue reports.
 - ▶ payment statistics.
 - ▶ report on suspense account activities; and
 - ▶ an audit trail of accesses and any adjustments made.

Table 8: Payments

Workstream (Level 0)	Functional (Level 1)	Area Functional Area Definition
Payments	Inquiry through walk-in	The ability for the taxpayer to opt for different services through the ISTD tax desk
	Making Payments	The ability for the taxpayer to view outstanding liabilities, generate a TRON, making a payment and viewing status of payments made
	Penalty Calculations	The ability for the system to calculate late payment penalties

	Payments on account	The ability of the taxpayer to make advance payments for future liabilities
	Payment methods	The various payment options available.
Workstream (Level 0)	Functional (Level 1)	Area Functional Area Definition
	Payment allocation	The ability of the system to allocate payments received towards outstanding liabilities based on certain business rules
	Payments made on third party applications	The ability for the taxpayer to make payments on any 3rd party applications like customs and have this data fed back to the system
	Payment history	The ability for taxpayers to view a history of all transaction on their account and print past payment receipts
	Outgoing payments	The ability of the system to process the payments for approved refund applications
	Requests from taxpayers	The ability for the taxpayer to raise requests to the ISTD via the system for issues like payment corrections, payment extensions, penalty reversals, conversion to instalment plans, etc.
	Notifications and correspondence	The ability of the system to automatically issue notifications to taxpayers and tax officers based on certain triggers

2.5. Compliance and Audit

During return processing, if any return is found deficient or have any mismatches or deficiencies, the related return may be selected for tax scrutiny assessment. Provision to have risk-based audit criteria as to focus on the cases with high probability of non-compliance. System should also be able to track timely payment of dues and impose financial sanctions (i.e., penalties and interest) for acts of noncompliance and to remit such sanctions in appropriate circumstances. Key features are:

- ▶ Conduct financial analysis of form (declaration/return) and other data to automatically select cases for audit.
- ▶ Electronic scrutiny of returns for assessment.
- ▶ Prioritize selected audit cases based on predetermined risk management criteria.

- ▶ Allocate cases for manual action via the case management system.
- ▶ Provide information for the preparation of the annual audit work plan.
- ▶ Provide tools to assist with audit activities, e.g., automation tools that can analyze taxpayer accounting records, links to other 3rd party information and reference material, and the ability to work remotely from a taxpayer site (e-inspection).
- ▶ Record audit activities and results.
- ▶ Update risk data according to outcomes of audit.
- ▶ Computation of penalties in the system
- ▶ Generate determined management information on the numbers, timeliness and results of audits, the success rate of audit cases selected, changes to selection criteria, and maintain an audit trail of any accesses or adjustments made.

Table 9: Compliance and Audit

Workstream (Level 0)	Functional (Level 1)	Area	Functional Area Definition
Compliance and Audit	Risk Engine		The ability to create and manage risk strategies to categorize taxpayers as high, medium, and low risk for various tax types and across various criteria such as sector, turnover etc.
	Audit Plan		The ability to create and manage audit plans containing individual taxpayers and creating audit cases from these audit plans.
	Performing Audits		The ability for the system to facilitate the process of performing audits by creating audit cases, allocating resources to audits, create working papers, prepare tax assessments, and audit reports.
	Penalty Calculation		The ability for the system to auto calculate penalties based on tax assessments amounts and, on the violations, discovered by tax officers during audits and create multiple tax assessment scenarios to calculate penalties for each scenario.

2.6. Enforcement

Authority to exercise, without referral to another body, certain enforcement powers associated with administration of laws (e.g., to obtain information from taxpayers and third parties and to impose lien over property in respect of unpaid debts) using business intelligence and analytics tools. Key features are:

- ▶ Shift from being an enforcing first organization to a service first organization
- ▶ Adoption of electronic notices
- ▶ Automated escalation of risky non-filers for enforcement activities
- ▶ Prompt & workflow-based write-off process
- ▶ Detect cases where there is a debt outstanding and payable.

- ▶ Automatically generate a demand for payment of the debt after a nominated period when the payment was not received, and payment of penalties and interest if the payment has been made late.
- ▶ Report all outstanding debts by taxpayer, so that coordinated action can be taken across all debts owed by the taxpayer in a “whole of taxpayer” view.
- ▶ After a further period, automatically issue a stronger demand for the debt to be paid.
- ▶ Automatic retention of a taxpayer’s debt history must be updated to the database.
- ▶ After a further determined period, if the debt is still outstanding, allocate the case for manual follow up via the case management system.
- ▶ Rank debt cases for manual intervention based on risk assessment criteria, e.g., size of debt, age of debt, number of revenue types involved, taxpayer history, etc.
- ▶ Support collection of outstanding debts by installments.
- ▶ Allow debt cases to be cleared via approved write-off processes.
- ▶ Support the compilation of local and national debt collection plans.
- ▶ Generate management information on level and composition of debt, the volume of new debt, the amount of debt collected, and the amount written off, the age of debt, the number of taxpayers who are in debt, status of cases allocated via the case management system, and an audit trail of any accesses or adjustments made.

Table 10: Enforcement

Workstream (Level 0)	Functional (Level 1)	Area	Functional Area Definition
Enforcement	Inspection Planning		The ability for the system to facilitate the process of annual inspection planning by managing inspection strategies and providing insights to help accurately create an inspection plan.
	Execute inspections		The ability for the system to facilitate the process of executing inspections by assisting in resource management, creating inspection cases, creation of working papers, creating inspection reports with violations, and making recommendations to other departments
	Asset Research		The ability for the system to look up information on taxpayer entity assets using internal and external data sources

	Risk Engine	The ability to create and manage risk strategies to categorize taxpayers as high, medium, and low risk for various tax types and across various criteria such as sector, turnover etc.
	Manage Collections	The ability for the system to support the debt collection related activities of the ISTD
	Tax Crime	The ability of the system to allow the enforcement department to recommend cases to external agencies for potential criminal investigation
	Manage non-filers	The ability of the system to support the enforcement department in managing non-filers of tax returns and levying estimated assessments

2.7. Legal, Appeals and Courts

System should provide the functionality for managing activities associated with resolving taxpayers' objections and appeals concerning adjusted assessments and rulings. Provision of the system to cover the processes related filing of lawsuit against taxpayer or prepare response against lawsuit received from court. Key features are:

- ▶ E-registration of appeals
- ▶ Time bound closure of appeal cases & introduction of appeal registration fees Pre-filled online tax appeal form
- ▶ Online dispute status tracking system: Receive and record taxpayer disputes, appeals, objections, and amendments which are then managed via the case management system.
- ▶ Adoption of paper-free workflows
- ▶ Automated case management system
- ▶ Generate determined management reporting with an automated audit trail of any accesses or adjustments made

Table 11: Legal, Appeals and Courts

Workstream (Level 0)	Functional (Level 1)	Area Functional Area Definition
Legal, Appeals and Courts	Manage Legal requests	The ability for Legal Department to respond to various requests regarding legal matters from various departments within ISTD

	Manage Taxpayer Appeals	The ability for Taxpayer to appeal to any decision by ISTD and possible changes to the original decision through an internal investigation by a committee formed by ISTD
	Dispute Management in Courts	The ability for Taxpayer to appeal outside the ISTD in the court of law to challenge a decision by appeal committee formed by ISTD

2.8. Information and Deductions

The function covers the granting of Tax Exemptions and thereafter their follow up process for check on Tax evasion and fraud. Key features are:

- ▶ Online Submission and procession of request for grant of tax exemption
- ▶ Online notification of decision to taxpayer
- ▶ Data-driven decision making
- ▶ Automated case management system
- ▶ Generate determined management reporting with an automated audit trail of any accesses or adjustments made

Table 12: Information and Deductions

Workstream (Level 0)	Functional (Level 1)	Area	Functional Area Definition
Information and Deductions	Tax Exemption Management		The ability for the system to manage request from users for tax exemptions/ purchase mechanism for tax type following defined workflow
	Golden Taxpayer Status		The ability for the system to manage request from users applying for Golden Taxpayer Status
	Preliminary Audit Review		The ability for the system to create and manage Audit Case following defined workflow, based on Return Submissions and perform automatic checks.

2.9. Customer Relationship Management

Provision of system to cover the interaction with various other functions and will keep updating itself as per the outcomes across other functions. Key features are:

- ▶ Electronic processes
- ▶ Adoption of paper free workflows

- ▶ Data-driven decision making
- ▶ Automated case management system
- ▶ Provide staff and taxpayers with access to rulings database, public information, standard questions, and answers for frequently asked queries.
- ▶ Ensure information and downloadable forms on the department's website are accurately maintained.
- ▶ Allow interaction with the taxpayer through the website, and with proper security, allow taxpayers to view their account details on-line.
- ▶ Provide channel choices for communication— online (web, mobile), email, text/SMS, phone, WhatsApp, walk-in, etc.
- ▶ Receive and record taxpayer correspondence which is then managed via the case management system.
- ▶ Generate determined management reporting with an automated audit trail of any accesses or adjustments made.

2.10. Refunds

Provision of the system to manage forecasting, planning, and budgeting activities for ISTD. In case of excess payment of taxes, online application, and issuance of refunds to the taxpayers in a timely manner. Key features are:

- ▶ Online Monitoring of Revenue Collection
- ▶ Identification of Taxpayers involved in fraudulent transaction
- ▶ Electronic disbursements of refunds directly to the account
- ▶ Risk based approach for validation/processing of refunds. Issue any subsequent refund electronically to a taxpayer's bank account
- ▶ Time bound disbursal of refunds
- ▶ E-Monitoring of refunds for better revenue forecasting & planning

- ▶ Develop and maintain an account for all taxpayers.
- ▶ Account for all debits and credits for all tax types in a 'whole of client' view.
- ▶ Enable all tax types to be recorded in the same style of account.
- ▶ Allow proper identification of all information passed to it from the payment and returns processing systems.
- ▶ Calculate due dates for incoming transactions.
- ▶ Allow for amendments to be made via reassessments, account adjustments, transfers, etc.
- ▶ Allow online enquiry to all details of a taxpayer's account.
- ▶ Allow, with appropriate security, taxpayers to view their account via web access.
- ▶ Calculate and impose late/nonpayment penalties and charge interest if appropriate.
- ▶ Automatically adjust penalties and interest if an amendment to an assessment is made.
- ▶ Structure the account so that details can be clearly identified between tax, penalties, interest, while being able to present all the taxpayer's account balances across all tax types.
- ▶ Age debts.
- ▶ Archive, to an environment that can be restored if necessary, nil balance account details at the appropriate determined time.
- ▶ Generate a statement of account, in either paper or electronic form for a particular tax type or consolidated for all tax types.
- ▶ Offset credits and debits within a tax type and across tax types.
- ▶ Provide for a variety of accounting transactions, e.g., debits, credits, transfers, refunds, penalties, payments, adjustments, write offs, etc.
- ▶ Produce management information on overall account status, general ledger for the department and wider government, and an audit trail of any accesses or adjustments made.
- ▶ Report on all revenue assessed across all tax types, nationally, by office, by sector, and by industry and can analyze information at further levels if required.
- ▶ Enable real time reports to be generated by department officials.
- ▶ Track revenue collection against predetermined budgets for specific tax types and for the revenue.
- ▶ Create and maintain revenue forecasting models that can determine "what if" scenario planning outcomes for tax policy budget changes, and for determining the possible result of a specific compliance strategy intervention work activity.

Table 14: Refunds

Workstream (Level 0)	Functional (Level 1)	Area	Functional Area Definition
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Refunds	Requesting Refunds	The ability for the taxpayer to complete, submit, amend, and manage refunds for different tax types
	Processing Refunds	The ability for the system to validate the refund request and create work items for the appropriate tax officer or team. This also includes the ability of the system to allow the tax department to modify/update the business rules as and when the need arises, without any technical or base code changes
	Paying refunds	The ability of the system to support the payment of approved refunds to taxpayers through the financials affairs system, track the status of payments and record the details of all electronic fund transfers
	taxpayer accounting	The ability of the system to maintain a taxable person ledger and perform automatic updates to the ledger based on various triggers and transactions
	Reconciliation	The ability of the system to perform reconciliations of incoming and outgoing payments against liabilities as well as bank statements and reports from external entities like payment gateways
	Collections management	The ability of the system to provide visibility on all outstanding receivables for tax officers to take appropriate actions
	Instalment Plans	The ability of the system to manage and support instalment plans

Workstream (Level 0)	Functional (Level 1)	Area Functional Area Definition
	forecasting	The ability of the system to forecast revenue in terms of taxes, penalties, interest, service fees and any other sources of tax related revenue

2.11. Dashboard and Reporting

System based control and assessment of tax operations for accountability and transparency. Key features are:

- Access to multiple dashboards and system driven escalations

- ▶ Electronic Workflow system for tracking and management
- ▶ System based Variance Analysis
- ▶ Receive and assemble data from multiple sources (data other ministries such as Customs, Ministry of Trade etc.).
- ▶ Cater for multiple views of data with multi-indexing capability.
- ▶ Provide analysis tools for risk assessment, trend analysis, "what-if" scenario analysis, etc.
- ▶ Generate pre-defined reports.
- ▶ Respond to requests for ad-hoc reports and analysis

Table 15: Dashboard and Reporting

Workstream (Level 0)	Functional (Level 1)	Area Functional Area Definition
Dashboards and Reporting	Corresponding to each functional module	Refer ITAS Annexure A– ITAS Functional Requirements for details

2.12. BI and Analytics

ITAS shall receive, store and process data related to taxpayer registration, returns, tax payments, etc. Over a period, this data will be accumulated to provide insightful information required for analysis and compliance as well as demonstrate interesting patterns. It is desirable to deploy an appropriate BI system to analyze and correlate the data to generate reports in various forms, that could assist the policy makers in policy formulation and economic analysis.

Table 16: BI and Analytics

Workstream (Level 0)	Functional (Level 1)	Area Functional Area Definition
BI and Analytics	Registrations Management Returns and Declaration Taxpayer Services and Accounts (Payments) Compliance and Audit Legal, Appeals and Courts Enforcement Information and Deductions (Tax Exemption) Customer Relationship Management (CRM) Instalment, Collection and Refunds (Revenue Management) Internal Audit	The system shall receive, store and process data related to taxpayer registration, returns, tax payments, etc. Over a period, this data will be accumulated to provide insightful information required for analysis and compliance as well as demonstrate interesting patterns. It is desirable to deploy an appropriate BI system to analyze and correlate the data to generate reports in various forms, that could assist the policy makers in policy formulation and economic analysis. Following key activities are envisaged: 1. Data extraction, transformation, and loading 2. Dynamic/Ad-hoc reporting (with slicing, dicing and basic modelling capability) 3. Develop management, KPI and Analytics dashboards 4. Taxpayer risk profiling 5. Training and capacity building 6. Predictive Analytics Models

3. ITAS Technical Requirements

ISTD envisages ITAS to be a high quality, proven integrated customer centric system that will streamline tax processing and improve operational efficiency. In-addition to transactional and historical data which is always important, ITAS should enable ISTD to also access and extract data in real-time to gain insights into current operations and improve decision-making.

The sections below provide an overview of each technical area in ITAS. Refer to the detailed technical requirements in the ITAS Annexure B - ITAS Technical Requirements. Bidder is expected to update the 'requirement compliance' column of this annexure as part of response submission.

3.1. ITAS Solution Architecture

ITAS needs to feature an enterprise architecture that supports the use of modular technology. ITAS must be based on modern technologies, including scalability from both a platform and architecture perspective. ITAS should support multiple user interface touchpoints while ensuring compliance with security standards and ease of use. ISTD prefers a COTS System in-order to minimize customization and avoid incremental cost for future system upgrades and ongoing maintenance. ITAS should be configurable beyond the core code to allow for changes, including tax rates and forms, on an ongoing basis.

The key design drivers when conceptualizing the ITAS solution are as below:

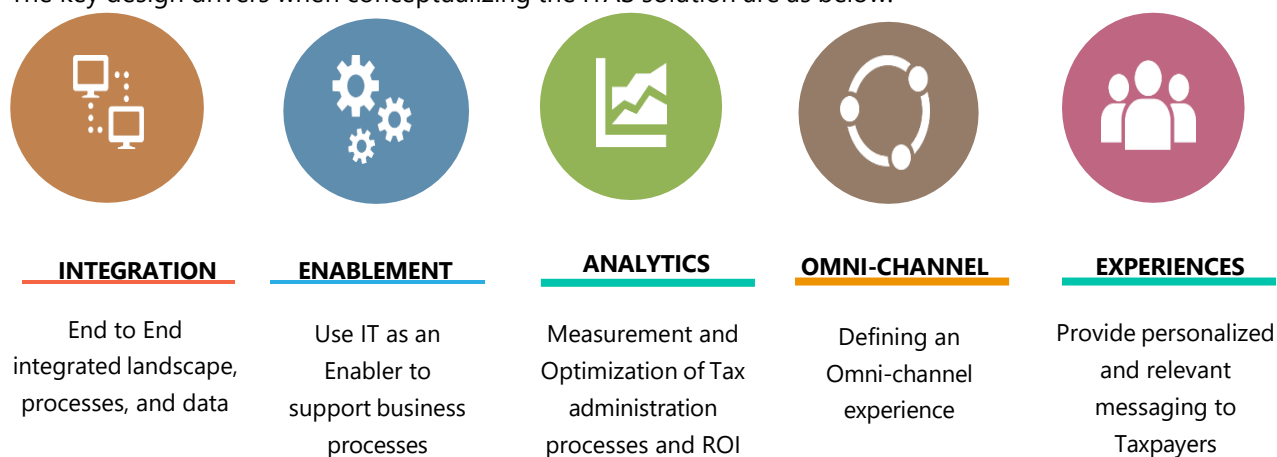


Figure 6: ITAS Solution Architecture

3.1.1. Platform Approach

The ITAS will be designed as a faceless system with API-driven architecture at its core. This will enable the access of all system features via any user interface (internal or third-party applications such as other government applications – Refer ITAS Annexure C - Interfacing Systems) that work on top of these APIs. The ITAS portal and mobile app will be applications on top of these APIs, rather than being fused into the platform as a monolithic system. One key benefit is the ability to change the technology of the portal/mobile app without impacting the core system. The core design mechanism ensures openness, a multi-user ecosystem, specific vendor/system independence and most importantly provides stakeholders with the option of using innovative applications on

various devices (mobile, tablets, etc.) that are built on top of these APIs. An ecosystem shall be built above the platform.

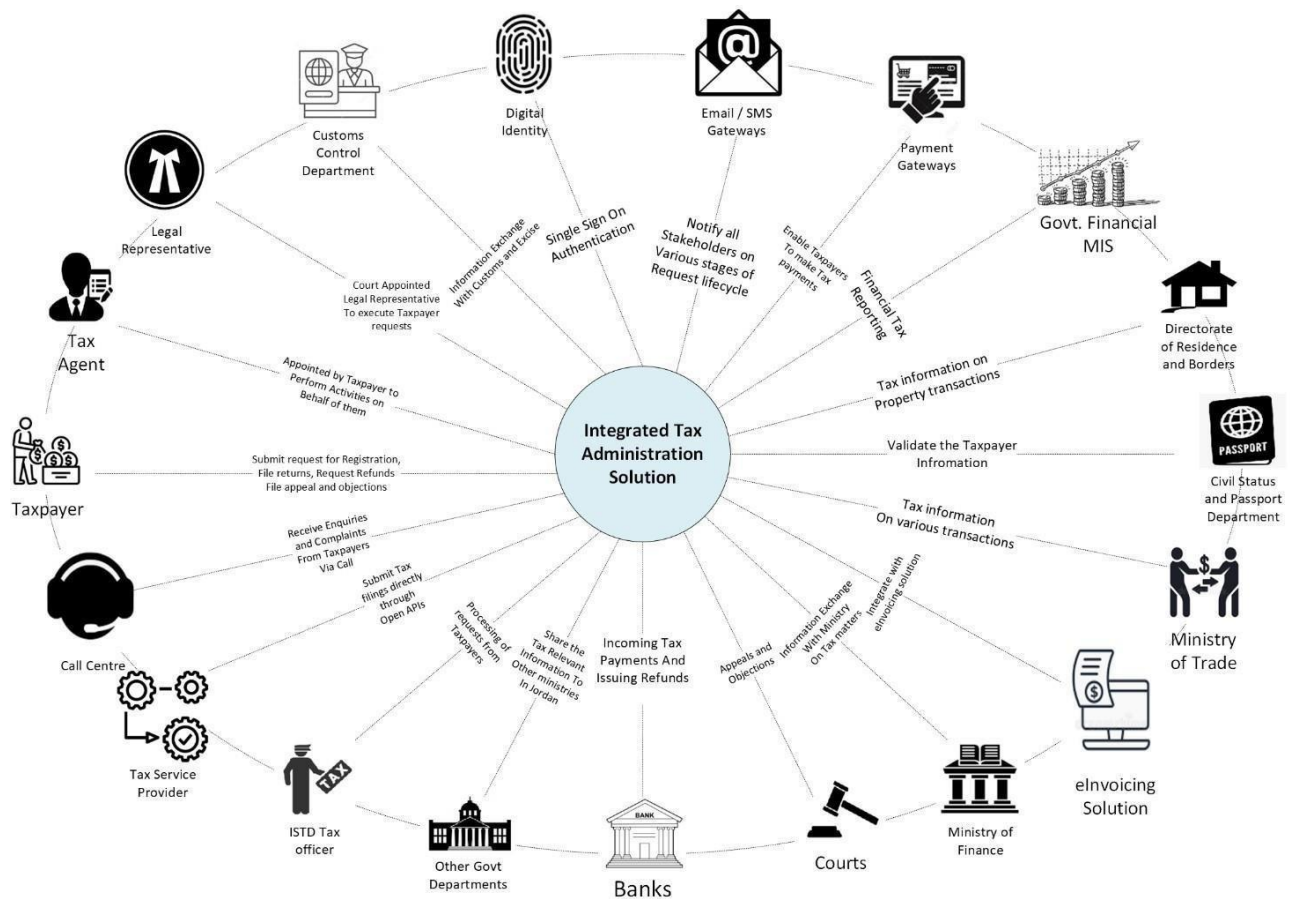


Figure 7: ITAS Ecosystem

3.1.2. Design Principles

3.1.2.1. Architecture Principles

The key architecture principles are listed below:

- ▶ Architecture should be highly flexible to support incremental upgrades in functionality across various services and functions delivered by the system. It should be able to accommodate the changing business environment within ISTD.
- ▶ Architecture should have a highly modular design to enable addition / modification / suspension of new modules and functionalities.
- ▶ Bidder shall propose Service Oriented Architecture (SOA)/Microservices based architecture while designing the solution.
- ▶ The system should be highly accessible and available through various devices across multiple locations.
- ▶ The principle of architecture should ensure keep data loss to Zero.
- ▶ The architecture should be cloud ready and should support interoperability through cloud platforms.
- ▶ Each input should be formed as a unique case in the system.
- ▶ There should be no single point of failure in the system.
- ▶ Bidder must maintain principle of first time-right data capture. The system should provide maker-checker functionality for all applications. System should allow multiple levels of checkers and approvals which may be configured within the system.
- ▶ Forms should provide client-side validation of inputs.
- ▶ All transactions and user activity being carried out within the system should be logged along with timestamp and client machine details in a format such they are searchable and allow for investigation as required. These should be recorded for future purposes for a defined period. In case of any judicial proceedings, the records should be maintained as long as the proceedings.

- ▶The system must be reliable against software failures. Integrity, confidentiality, and availability of data must be assured.
- ▶System should provide for seamless integration with various internal and external systems through required interfaces. The unique identity of the user should be maintained across the system with a single sign-on facility. This facility should be implemented without bypassing any security principles.
- ▶The underlying technology needs to be user friendly by having ease of use principle. By having an ease of-use principle, training can be kept to a minimum thereby aiding IT change management and the risk of using a system improperly can be minimized.
- ▶Training can be kept frequently thereby aiding IT change management and the risk of using a system improperly can be minimized.
- ▶Vendor lock-in should be avoided.
- ▶The bidder should document all the integration efforts of the proposed solution with Interfacing Systems (To-Be) listed in the ITAS Annexure C - Interfacing Systems.
- ▶API based approach: Open standard based APIs will support data exchange mechanisms between third party systems.
- ▶Interoperability: The system is expected to be interoperable. This is the key to sharing resources and reduction in long-term development costs.
- ▶ N – Tier Model: N-Tier model is the framework in which application user interface, logic, data, and their associated processing and tiers are separated from each other in logical manner. This isolates/minimizes the impact of change and It is easier to support, is more scalable and supports interoperability.
- ▶Openness: Open API, open standards should be considered for designing the system. The bidder should consider scalability, security and performance of the system while deciding on the system and adherence to the open architecture design principle.
- ▶Reliability: The system must have appropriate measures to ensure processing reliability for the data received or accessed through the solution.
- ▶Design, develop, implement, deploy (install, test, launch) and rollout of the proposed solution This needs to be aligned with the e-Government Architecture Framework including the use of shared components and services like the SMS Gateway, Central Bank Payment Gateway eFAWATEERcom, e- Government Contact Center, Government Service Bus (GSB)

- ▶ Develop scanning functionality/feature on the walk-in application to enable Entity users to scan documents and upload it to the system, the application should be integrated to the Entity archiving system.
- ▶ Provide very well documentation for the built APIs.
- ▶ Working closely with MODEE – Datacenter teams to deploy the built APIs over GSB and test it
- ▶ Provide the Services on SANAD Platform as Native services using the APIs.
- ▶ Build the needed template for providing the services through SANAD Platform.

3.1.2.2. Security Principles

The key security principles are listed below:

- ▶ The bidder should follow security principles such as “Defense in Depth with Zero trust mechanism”, a multi-layer defense mechanism, so that an attacker must defeat multiple mechanisms to perform a successful attack.
- ▶ Bidder must ensure the security architecture model to facilitate effective incident response resolution, forensic investigation during incident analysis with best practices like real time internal network defense, etc.
- ▶ Bidder must be aware of threat and its mitigation for the application which include spoofing, tempering, repudiation, information disclosure, denied of service and elevation of privilege along with OWASP (Open web/Mobile application security project) testing guidelines.
- ▶ The system should be compliant to guidelines for Web /Mobile/ Application / Network Security. The System should provide the date and time of all transactions with details of creation, read, update, delete or print. Access should be restricted at different levels of data file, program, module, screen, record, field database table, row, or column.
- ▶ All the Applications and Infra changes must be by secure SMLC (Security Management Life Cycle) and Change Management principle driven respectively.
- ▶ The bidder should follow the principle of “least privilege”. Each user and program should operate using the fewest privileges possible. The bidder will have to design its solution utilizing similar industry recognized security principles.
- ▶ All IT operations will be governed by the IT Policy which will be provided to the successful bidder. The bidder will have to prepare detailed procedures for the same and implement them accordingly. All project documentation should be prepared by the bidder as per the policy and related regulations.
- ▶ The privacy of data must always be ensured by the bidder. The bidder must ensure that data sharing is done as per the Jordan Data classification and management policy and Jordan personal data protection law.
- ▶ Bidder must adopt technical and administrative measure to protect personal data from loss, misuse or alteration based on global best practices for privacy and security such as OECD, Jordan Data classification and management policy, ITIL standards and Jordan personal data protection law

- ▶ The system must follow role-based access control at all levels. All the access logs need to be captured and monitored.
- ▶ Application Access should follow 2 Factor Authentication.

All databases and data stores must be encrypted. Safeguard data while at rest and in transit. Identify and classify the data in terms of criticality / sensitivity and define their levels.

- ▶ Bidder must ensure data security life cycle as a principle in securing data while creating, storing, sharing, archiving, or destroying.
- ▶ Bidder must ensure database protection with database activity monitoring and file activity monitoring.
- ▶ Bidder shall get the Vulnerability Assessment (VA) and Penetration Testing (PT) as security audit conducted by an empaneled agency before deployment/ Go-Live of project. The cost of which should be borne by the bidder.
- ▶ Security in design would encompass secure information architecture, proper role-based access design and secure application and database design.
- ▶ The bidder must ensure that their Application Development must follow secured SDLC process development and deployment by taking OWASP Top 10 and SANS top 25 into consideration. Similarly, Application maintenance should follow secured SMLC.
- ▶ The system must be secure at all user touch points by using suitable security protocols and data protection methods Industry best practices for coding of application to ensure sustenance to the Application Vulnerability Assessment.
- ▶ Audit trails and Audit logging mechanism to be built in the system to ensure that user action can be established, and investigations (if any) can be aided. (E.g., Logging of IP Address etc.)
- ▶ Data should be visible to authorized entity only. Data alterations etc. through unauthorized channels should be prevented.
- ▶ Appropriate authentication mechanism adhering to industry best practices regarding Password Policies etc.
- ▶ The bidder will ensure that the critical data stores are minimized, and stored data must be always encrypted.
- ▶ The bidder must ensure that access to data is always given through the application layer (via an application).
- ▶ As a part of Service delivery process, the bidder must ensure segregation of services and segregation of duties.
- ▶ For operations phase security activities such as performing backups, holding training classes, managing cryptographic keys, keeping up with user administration and access privileges, and updating security software are some examples that must be done by the bidder.
- ▶ VPN and VLANS should be the principle of operations for remote access and isolation of internal traffic.

- ▶ Web facing application should be deployed in DMZ (Demilitarized) zone and the Database Server should be deployed in the secured zone while deploying the application on Government Private Cloud.
- ▶ Bidders must adopt metrics to measure risk management performance. E.g., Cyber security information exchange framework, security content automation protocol etc.
- ▶ Monthly / Periodic reporting on security breaches and attempts plus the action taken to thwart the same and providing the same to ISTD.

Performing periodic review of security configurations for inconsistencies and redundancies against security policy.

- ▶ The bidder should embed a security incident response plan within the Business continuity and disaster recovery plan to response in an efficient and effective manner in case of a disaster.
- ▶ The bidder must provide a detailed threat model analysis report and validate the same is correctly implemented across environments.
- ▶ The bidder must ensure the connection strings are highly secured.
- ▶ Bidder shall ensure that all servers (web, application, database, etc.) related to the system is configure securely based on the best practices.

3.1.2.3. Management Principles

The key management principles are listed below:

- ▶ The management of the system shall be SLA based, through an automated SLA monitoring system.
- ▶ SLA management will take place via Enterprise Management System (EMS) which will provide system generated reports on level of compliance/non-compliance against defined SLA parameters.
- ▶ System should have a Security Information and Event Management (SIEM) solution that provides end-to-end, comprehensive, modular, and integrated management of IT Security components to maximize the availability of application services and SLA performance.
- ▶ Application Management shall follow all processes as per to ISO 20000/ ITIL standards. This includes Configuration Management, Incident Management, Performance Management and Capacity Management.
- ▶ Real time reporting to help management and administrators to take quick decisions.
- ▶ Application Security Management should intelligently perform root-cause analysis to rapidly bring the system back to normal working conditions.
- ▶ The system should be upgraded without affecting the production.
- ▶ It should be possible to proactively manage all software components maintenance and licenses throughout their life cycle.

3.1.3. High level solution architecture

ITAS is envisaged to be a layered, loosely coupled, Modular, Plug and Play, SOA based architecture ensuring Agility, Flexibility, Interoperability and Performance Reliability. A reference architecture for the ITAS is shown below. The bidder will however need to define its own architecture by meeting all the requirements listed in this document.

The solution Architecture of the ITAS would be multi layered. The architecture is scalable both vertically and horizontally with security features. The salient features of the layers are as follows:

- ▶ **Presentation Layer:** The user shall interact and transact business with this layer. The layer shall provide the features to users such as personalization, enterprise search, analytics etc. All services either requested or delivered would be orchestrated through this layer.

Web Experience and Content Management Layer: This layer of the ITAS provides for content targeting, content authoring, Web Content Personalization, Multi-Channel Delivery, Information Rights Management etc.
- ▶ **Process and Service Integration Layer:** This layer facilitates integration of multiple business processes with the help of defined business rules. It also facilitates real time dashboard and data exchange.
- ▶ **Application Layer:** This layer of the ITAS would have the various software applications which serve the business functionalities of the ISTD. Additional services shall be added, modified at this layer. The layer on top of this integrates the various business processes.
- ▶ **Infrastructure and Data Layer:** This layer contains the host of hardware, software, Network, used for storing data, exchanging data, system software, business software etc.
- ▶ **Security Confidentiality and Control Services:** This runs vertically across the entire architecture providing security services to the various services. The architecture provides security services such as Single Sign On, Authorization and Privacy, Secure Communications, Identity Management, Role Management etc.
- ▶ **External Interface:** This layer facilitates providing access to external stakeholders in pushing and pulling data with the ITAS.

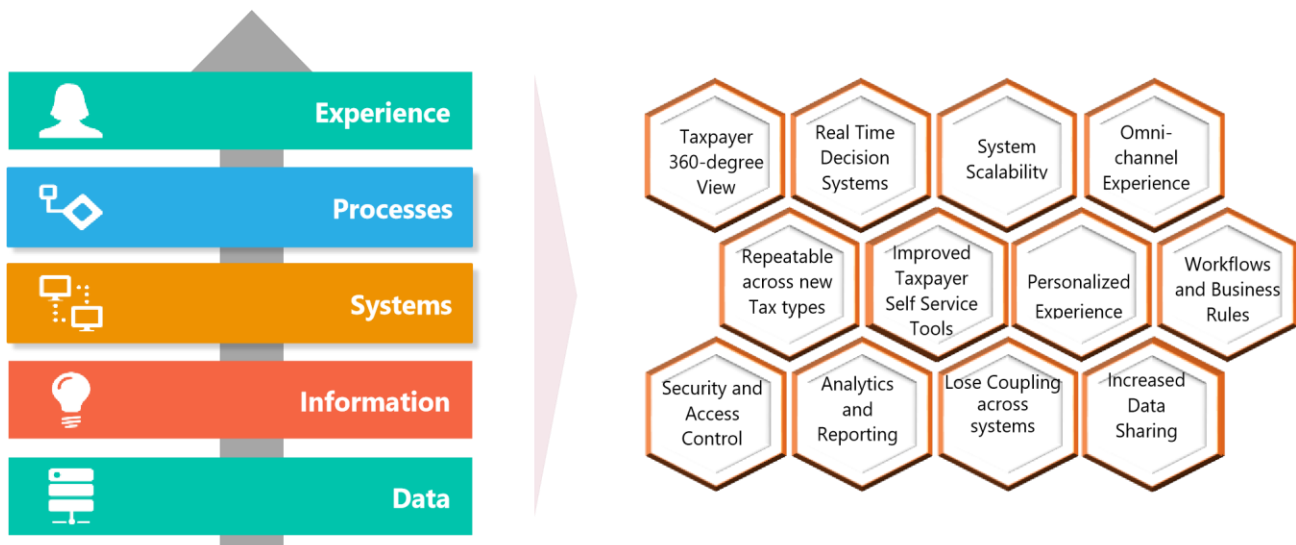


Figure 8: ITAS Layered solution and capabilities

Figure 8: ITAS Layered solution and capabilities.

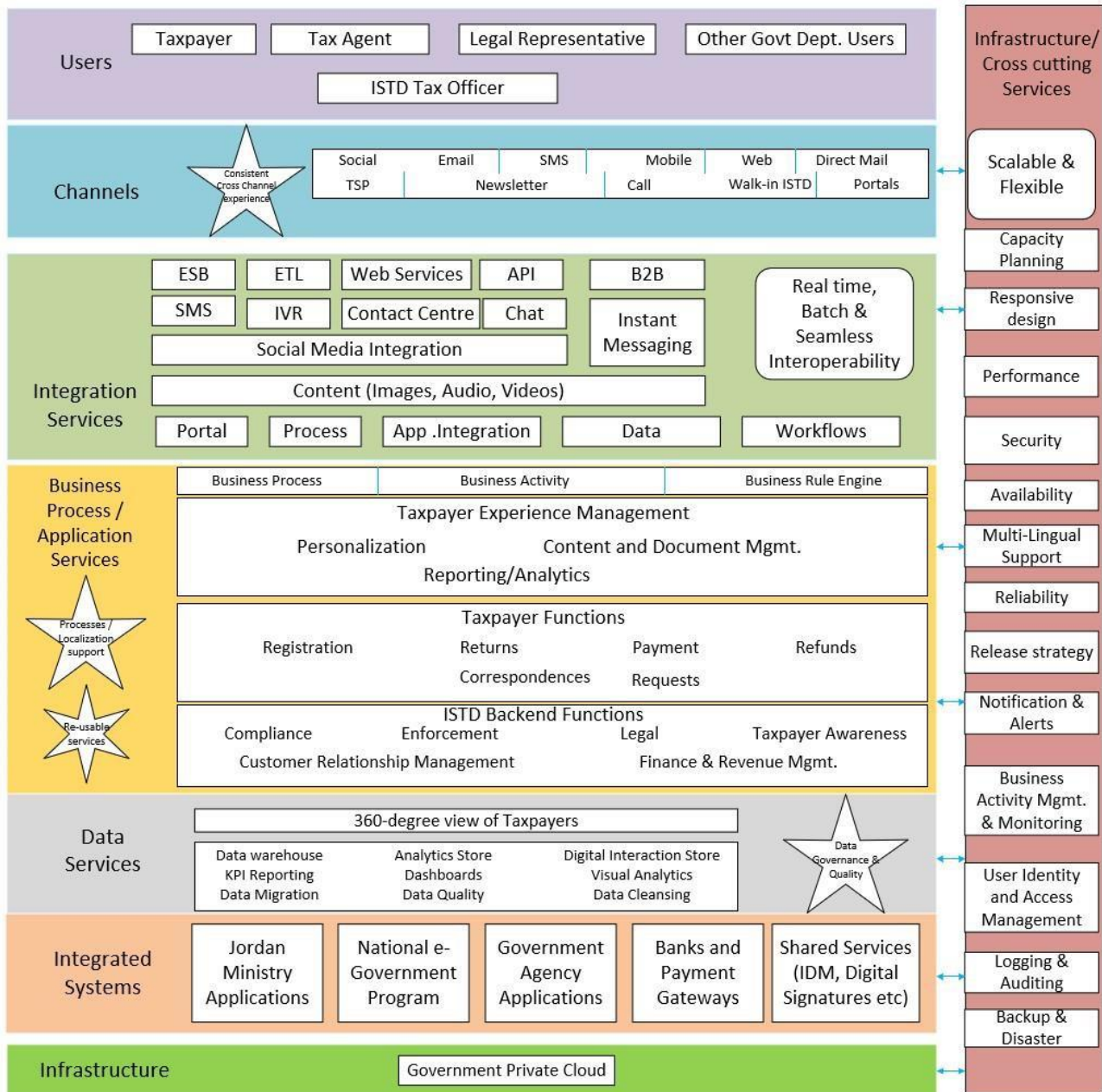


Figure 9: ITAS Reference Solution Architecture

3.2. Infrastructure

The winning bidder is required to perform the activities mentioned below regarding solution Infrastructure:

1. Must provide the solution hosted on the Government Private Cloud (GPC) and fully integrated with Nutanix AHV/VMware vSphere.
2. Must provide the solution design architecture that includes the required virtual machines (VMs), databases (DBs), or any other related services (GPC and infrastructure services are listed in the below table*).
3. Must describe each component's functionality and role in the architecture, knowing that GPC provides 3-tiers architecture for management and security purposes (Web, App, and Data).
4. Must provide the required sizing (computing specifications) that will be utilized for VMs and DBs.

* GPC provides the bidders with different services as shown in the below table.

Table 17: GPC Services

Government Private Cloud (GPC) Services		
1	Virtual Machines	
	Option 1	Windows Server 2022 Enterprise Edition (Licensed).
	Option 2	UPUNTU / Oracle Linux
	Option 3	Any other OS is the bidder responsibility to provide, install, configure, and license it. Any required licenses must be included in the financial proposal as optional item.
2	Databases	
	Option1	Installing your own MySQL , MS SQL server, Oracle DBs or any other DBs. In this case, it is the bidder responsibility to provide, install, configure, and the DB licenses .
3	Other services provided by modee	
	1.	Load Balancer (LB)
	2.	web application Firewall (WAF)
	3.	Publishing & DNS
	4.	Object storage
	5.	Micro Segmentation
	6.	Backup Solution. Winning Bidder must provide the backup policy
	7.	Warm DR Site on infrastructure level; the bidder must provide business continuity plan for all proposed solution components
	8.	Antivirus
	9.	SSL certificate
	10.	SMTP Integration
	11.	SMS Gateway integration

3.3. Service Level Agreement (SLA)

3.3.1. SLA management

The bidder should propose a design in compliance with service levels during the operations of the system. The SLA monitoring should not affect the basic functionalities of the component and should be available to ISTD in real time. The bidder shall be responsible for the monitoring and reporting of the SLAs on a periodic basis along with automated alerts on reaching predefined threshold values. The bidders can propose the usage of any tools to aid this process.

3.3.2. SLA change control mechanism

ISTD may amend this SLA by mutual agreement in accordance with the process described below:

1. Changes can be proposed by either party.
2. The Bidder will initiate an SLA review at least bi-annually. ISTD and Bidder will review the Service Levels and mutually agree whether to:
 - a. Add to, delete, or change the Services to be measured and the corresponding Service Levels to reflect changes in the ISTD's business operations; and
 - b. Improve the existing Service Levels, where warranted, to reflect operational or technical improvements.

3.3.3. SLA severity level

A maximum level of performance for Liquidated Damages are established and described below.

During operations phase while the SLAs measurement to be done monthly, the deductions for not meeting the SLAs shall be aggregated and will be deducted from last month's payment of a quarter.

- ▶ Performance Liquidated Damages shall be levied for not meeting each SLA.
- ▶ Breach of SLAs target shall result in various severity levels based on the business impact the component causes to Winning bidder services.
- ▶ If on performance evaluation it is found that the Winning bidder has not met a measurement parameter for two consecutive quarters the same shall result in doubling the Liquidated Damages percentage of that measurement parameter.
- ▶ The aggregate Liquidated damages for SLAs relating to Implementation Phase will be capped at 10% of the contract value of the agreement of Winning bidder.
- ▶ If the SLA penalties during operations Phase calculations exceed 10% of the monthly payment for three consecutive months, then ISTD may take appropriate action including termination of the contract and forfeiting of Performance Bank Guarantee
- ▶ SLA penalties for period during project initiation and until final Go Live shall be calculated on the figures of year 1 monthly payments.
- ▶ Notwithstanding anything contained herein, if the Winning bidder fails to achieve the Go-Live within 6 months of the planned date, due to reasons attributable to the winning bidder, then ISTD will have the right to terminate the contract for default.

The Winning bidder needs to supply and implement an integrated SLA monitoring tool as part of its solution which will monitor all the SLAs automatically and will provide automated report on monthly basis. The tool access to be available to ISTD on real time basis through a dashboard.

These design considerations are common and must be considered for all facets of design, development, and deployment of the proposed ITAS Solution.

3.3.4. System security SLAs

ITAS must be highly secured with “Zero trust” and “Defense in Depth/ Layered security” strategy should be guaranteed at each component of the system to confirm data integrity.

Defense in depth layers security restrictions should be able to adapt and adhere to the MoDEE/ISTD security guidelines.

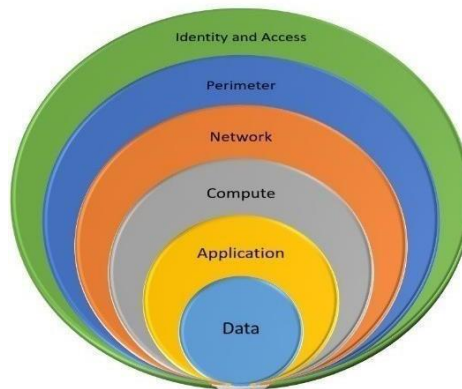


Figure 10: ITAS Security Layers

3.3.5. Operations Related Service Level

Table 19: Operations related SLAs

	SLA Parameter	Description	Target	Liquidated damages
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1.	Time to Resolve Severity 1	Time taken to resolve the reported problem.	For Severity 1, 100% of the incidents should be resolved within 60 minutes of problem reporting	Liquidated damages shall be levied as per the following table: <table><tr><td>% of incidents with more response time as mentioned in SLA table</td><td>Liquidated damages as % of the Monthly payments during Operations & Maintenance Phase</td></tr><tr><td><95% & >=93%</td><td>1%</td></tr><tr><td>< 93% & >= 91%</td><td>2%</td></tr><tr><td>< 91% & >= 89%</td><td>4%</td></tr></table> For each additional drop of 1% in performance below 89%, 2% of Monthly payment Operations & Maintenance cost shall be levied as additional liquidated damages.	% of incidents with more response time as mentioned in SLA table	Liquidated damages as % of the Monthly payments during Operations & Maintenance Phase	<95% & >=93%	1%	< 93% & >= 91%	2%	< 91% & >= 89%	4%
% of incidents with more response time as mentioned in SLA table	Liquidated damages as % of the Monthly payments during Operations & Maintenance Phase											
<95% & >=93%	1%											
< 93% & >= 91%	2%											
< 91% & >= 89%	4%											
2.	Time to resolve – Severity 2 & 3	Time taken to resolve the reported problem.	100% of Severity 2 within 4 hours of problem reporting Severity 3 within one working day of problem reporting	Liquidated damages will be levied as per the following table: <table><tr><td>% of incidents with more response time as mentioned in SLA table (Severity wise)</td><td>Liquidated damages as % of the Monthly payments during Operations & Maintenance Phase</td></tr><tr><td><95% & >=93%</td><td>0.25%</td></tr><tr><td>< 93% & >= 91%</td><td>0.5%</td></tr><tr><td>< 91% & >= 89%</td><td>1%</td></tr></table> For each additional drop of 1% in performance below 89%, 1% of Monthly payment of Operations & Maintenance cost shall be levied as additional liquidated damages.	% of incidents with more response time as mentioned in SLA table (Severity wise)	Liquidated damages as % of the Monthly payments during Operations & Maintenance Phase	<95% & >=93%	0.25%	< 93% & >= 91%	0.5%	< 91% & >= 89%	1%
% of incidents with more response time as mentioned in SLA table (Severity wise)	Liquidated damages as % of the Monthly payments during Operations & Maintenance Phase											
<95% & >=93%	0.25%											
< 93% & >= 91%	0.5%											
< 91% & >= 89%	1%											

3.	Percentage of reopened incidents	For all incidents that are marked as Resolved by the Service Provider, but are reopened by the client. This is calculated for all incidents	<= 2%	Liquidated damages shall be levied as per the following table:	
				% of reopened incidents	Liquidated damages as % of the Monthly payments during Operations & Maintenance Phase
				<=4% & >2%	0.5%
				<=6% & >4%	1%
				<=8% & >6%	2%
For each additional increase of 2% in re-opened incidents above 8%, 2% of Monthly payment of Operations & Maintenance cost shall be levied as additional liquidated damages					

3.3.6. Severity Definitions

All the issues on logging shall be assigned a severity level. Basic guidelines for assigning severity are as follows:

Table 20: Severity Definitions

Severity	Definition
Severity 1	Severity 1 problems are the ones which have a critical business impact. These problems will have any of the following characteristics: • Complete or part of any service unavailable (including APIs) • Incorrect behavior of the system
Severity 2	• Security Incidents • Data theft/loss/corruption • Severe impact on customer satisfaction/MODEE reputation – bad media publicity • No work-around to mitigate the disruption in service • Repeat calls (same problem that has occurred earlier reported more than 2 times) • Financial impact on ISTD Severity 2 problems are the ones which have a significant business impact. These problems will have any of the following characteristics: • The efficiency of users is impacted • Has a viable workaround • Severely degraded performance (slow service)

Severity 3	Severity 3 problems are the ones which have a minimal business impact. These problems will have any of the following characteristics: • No impact on processing of normal business activities • A low impact on the efficiency of users • Has a simple workaround • Enhancement requests
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3.3.7. Cross Cutting Services

3.3.7.1. User Identity and Access Management

The ITAS should integrate with IDM SSO. The System should also support multi-factor authentication. System should also be integrated to Jordan's Digital Identity Management (IDM).

3.3.7.2. Redundancy

The bidder must provide a detailed redundancy architecture diagram and strategy to eliminate single point of failure for all the components of the system which helps in assuring the availability, RPO and RTO SLA

3.3.7.3. Replication

The bidder must provide a detailed replication architecture diagram and strategy to achieve the RPO and RTO SLA

3.3.7.4. Archiving

The bidder must provide a detailed archiving and retrieval architecture diagram and strategy to achieve performance and scalability requirements. Refer to ITAS Annexure D – Data Points for ITAS.

3.3.7.5. Capacity Planning

Servers must be supplied with all systems hardware modules and software tools that are required to operate the servers in a hot-standby highly available configuration. The solution should provide load balancing, scalability, and performance enhancement through added cluster nodes besides high availability. All hardware should have a clear and easy upgrade-option in case of performance bottlenecks by using the scalability options. Bidders are also required to ensure that the proposed hardware meet the performance and sizing requirements of ITAS considering the peak loads.

3.3.7.6. Responsive design

The various channels to access the ITAS (Front end and back end) should be responsive to be accessed from multiple devices or varying sizes and resolutions through all the leading browsers and operating systems

3.3.7.7. Performance

The ITAS should have high performance for all the components and should be independent of each other. The ITAS should have minimum downtime and system response time considering the peak loads too.

3.3.7.8. Availability

The ITAS must be available 24 / 7 of operations and support domestic Jordan based operations. The maintenance should be planned during off business hours, weekends and as per the holiday calendar of ISTD.

3.3.7.9. Multilingual Support

The ITAS should have multilingual support in English and Arabic through all the channels such as web portal, Mobile, Call, Social Media for all the user personas including the Taxpayers and ISTD Tax officers.

3.3.7.10. Reliability

The ITAS should be designed to be reliable to handle confidential and sensitive information related to Tax functions for the Taxpayers and the ISTD back-end functions complying to the necessary security requirements. The ITAS should maintain the data integrity and prevent data loss and data redundancy.

3.3.7.11. Release Strategy

The bidder should describe the release plan and strategy for the below aspects but not be limited to the management of future version releases, service packs, patches, functional and/or technical upgrades, configuration, backup/restore, Enhancements, and Error and defect resolution.

3.3.7.12. Notifications and Alerts

The ITAS should notify the various personas including the ISTD admin, Tax officers, Taxpayers on various alerts, reminders, notifications, correspondences, and decision based on configurable predefined triggers. ITAS can also extend the modes of notifications to emails and SMSs based on business process requirement.

3.3.7.13. Business Activity Monitoring and Management

The ITAS should have a database Real-time protection and security technology for auditing, monitoring, and analyzing activities. The Bidder should describe the process of Monitoring of various aspects like but not limited to database, system availability and performance SLAs, business activities, transactions, load capacity and alert the respective ISTD personnel on the same.

3.3.7.14. Logging and Auditing

The ITAS should have the ability to access security logs and reports relating to the System, including, at minimum, latency statistics, user access summaries, user access IP address summaries, and user access history and security logs for all for all the user personas to the front end and back-end interfaces.

The ITAS should also have the ability to perform security audits/scans of the System, including the hosting environment, including unannounced penetration and security tests, and how ISTD can incorporate recommendations that result from such inspections, tests, and audits as part of its security/compliance program, including standard remediation timeframe.

3.3.7.15. Backup and Disaster Recovery

The Bidder should describe ITAS's Disaster Recovery and Business Continuity Plan, including but not limited to:

- ▶ Redundancy, Data Backup, and Disaster Recovery
- ▶ Data Export/Import
- ▶ Data Ownership and Access
- ▶ Testing

- Assumptions or recommendations for business process recovery time objectives (RTO) and data recovery point objectives (RPO)

3.3.7.16. System Administration

The system should contain administration module, to enable administrators to perform all day-to-day administrative tasks at data, business rules engine, and application levels.

The Winning bidder should gather all ITAS related administration requirements during business requirements gathering and analysis phase.

System should allow admin to perform the following tasks but not limited as follow:

- Manage user profile.
- Manage Security Permissions.
- Manage lookups

3.4. Security

Security of system and exchanged transaction information should be guaranteed at all system layers based on WS-Security standards including infrastructure, application, web services and integration points, and access channels. This also includes using detective and preventive controls for all security threats and approval by ISTD and MoDEE. A suggested Information security management framework is shown below.

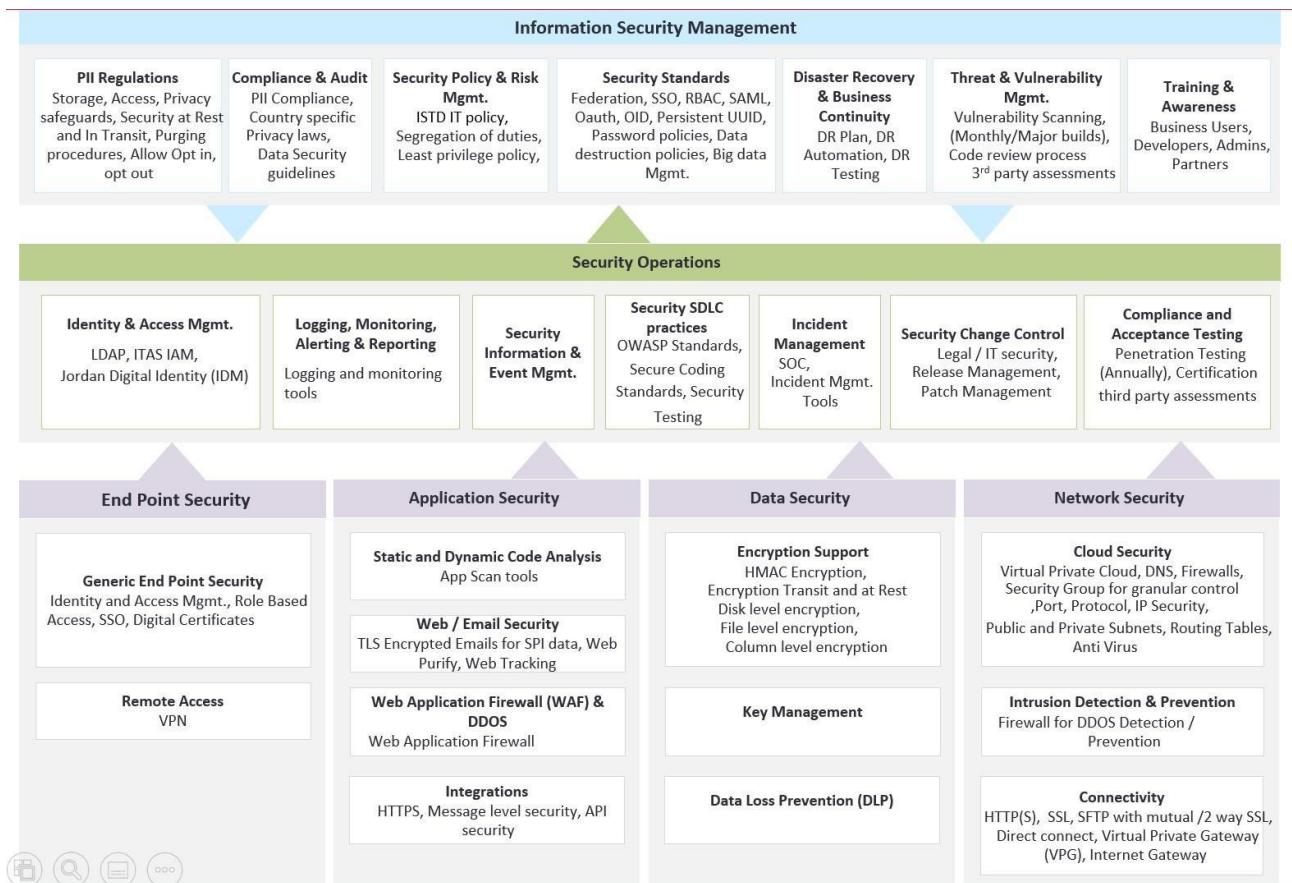


Figure 11: ITAS Information Security Management framework

Bidder's description should encompass, at a minimum, the items outlined in the below section. The bidder will

also be responsible for participating in periodic audits to ensure that compliance is successfully maintained by the bidder over the duration of the Program. The bidder is expected to maintain compliance and participate in audits as defined below at no additional cost to the ISTD.

System compliance

Bidder should describe its general approach to information security and privacy compliance as per the above information security management framework, and the general method by which compliance is addressed and achieved. Bidder should describe any Third-Party certificates of audit certifying on a recurring basis that the Application Services and System(s) comply with. Bidder should describe any other Third-Party certifications, tests, or reports the bidder obtains on a recurring basis and that it will provide to the ISTD at no cost upon request, including but not limited to any of the following:

1. Latest ISO 27001 certification
2. Test or assessment results of an independent, Third Party application scans using the Open Web Application Security Project (OWASP) Top Ten List, SANS 25 list
3. Test results of a penetration test of the system(s) conducted by an independent Third Party (reviewed and approved by MODEE)
4. Bidder's operations, information security program, and disaster recovery/business continuity plan

Respondent should provide the latest remediation plans which may have resulted from any compliance activities or efforts listed above.

Security audit

The bidder should describe the mechanism to perform security audits/scans of the System, including the hosting environment, including unannounced penetration and security tests, and how the bidder will incorporate recommendations that result from such inspections, tests, and audits as part of its security/compliance program, including standard remediation timeframes.

Access to security logs and reports

Describe the mechanism to access security logs and reports relating to the System, including, at minimum, latency statistics, user access summaries, user access IP address summaries, and user access history and security logs. Provide samples of standard security logs and reports.

3.5. Data Migration

Data Migration in the context of this RFP refers to validation and migration of data from the current eServices system to the new database schema, linking and Meta tagging the documents to the relevant records in the Data Management System (DMS) and conformance to quality control requirements. The Data points related to the current eServices system are detailed in the ITAS Annexure D - Data points for ITAS. Data should be migrated from the current application and media to the new application prior to 'go-live' of the application.

Data Migration comprises the process for managing the data requirements for the System, including data extraction, conversion, cleansing, migration and transition methodology and approach. The ISTD also expects the bidders to include, as part of their conversion approach, a description on how it will work with ISTD Staff to come to an agreement on various items, including but not limited to volume of data to be converted, level of detail, data objects, what data should be made available in an archived "read-only" state, etc.

The Bidder shall also highlight the proposed tools and framework to be used for seamless transition from ISTD's current e-Services and core tax system to the proposed ITAS solution. The Bidder shall propose a solution for digitization of existing documents covering document taxonomy, metadata, classification, scanning (or conversion into soft copy), archival, storage and retrieval / readability from various structured, unstructured, and physical data sources. Since the data is also in Arabic, the bidder shall deploy requisite experts with Arabic knowledge. In order to effectively perform data assessment and cleaning, it is important that the data migration team is sufficiently familiar with existing data structures of the source registers/databases.

The Bidder should describe their data migration strategy, approach, and methodology by considering key factors including historical data, open transactions, and backlogs. The bidder should define the process which is to ensure the successful and timely conversion of data from the existing legacy system into the new proposed system. The bidders need to provide: Data Migration Design, Data Migration Execution Plan and Cutover Execution Plan. All results of the data migration test (in lower environments), and the actual migration must be preserved and accessible for future use.

The approach should include the following:

- ▶ A data conversion approach including blueprint document and plan.
- ▶ Identification of any risks associated with data conversion, cleansing and conversion.
- ▶ An approach and plan for data conversion testing. The data migration test and execution process should follow the ISTD security policy and security requirements
- ▶ Mock data conversions and their alignment to testing activities
- ▶ Data reconciliation reports including remediation of data to be carried out by ISTD prior migration
- ▶ Data governance and versioning.
- ▶ Data conversion mapping and transformation.
- ▶ Data conversion signoff and cutover processes.

A high-level migration project phases is shown below for consideration:

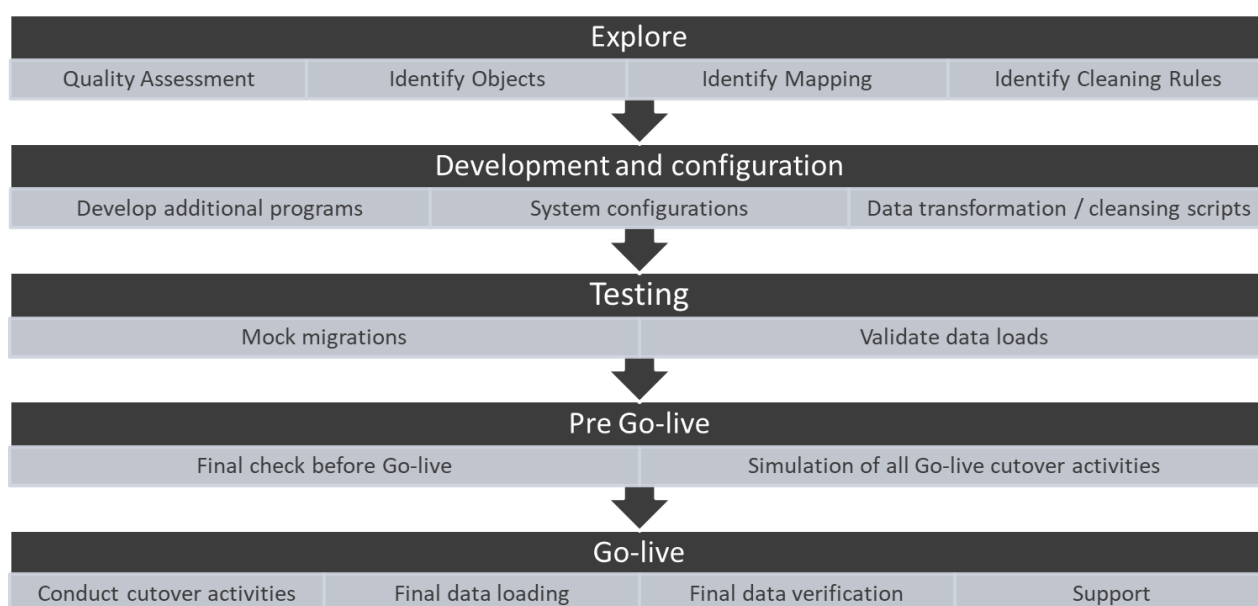


Figure 12: Data Migration

3.6. Reporting and Analytics

3.6.1. 360-degree view of taxpayer

The ITAS will use the CRM for a 360-degree view of the taxpayer. The CRM system will provide a well-defined platform for ITAS to interact with taxpayers and fulfill their needs and demands in an effective manner. The bidders need to provide the approach towards CRM implementation and CRM architecture as part of their proposal.

3.6.2. MIS, Reporting, & Business Intelligence (BI)

The BI component is mostly dedicated for the ITAS internal consumption. The BI and analytics platform would consist of the following components:

1. Data-warehousing and analytics
2. Visualization and reporting
3. Analytics & Digital Interaction Store: Over time, the data being collected and stored by the ITAS will accumulate to provide a wealth of information and insights. These contribute to useful analysis for compliance as well as to demonstrate interesting patterns and exceptions. It is desirable to make use of an appropriate Business Intelligence tool to analyze and correlate the data and generate reports in various forms. This would provide necessary inputs and enable the ISTD to derive useful insights and information. These insights and information can further be utilized to enhance business rules/processes within the ITAS.

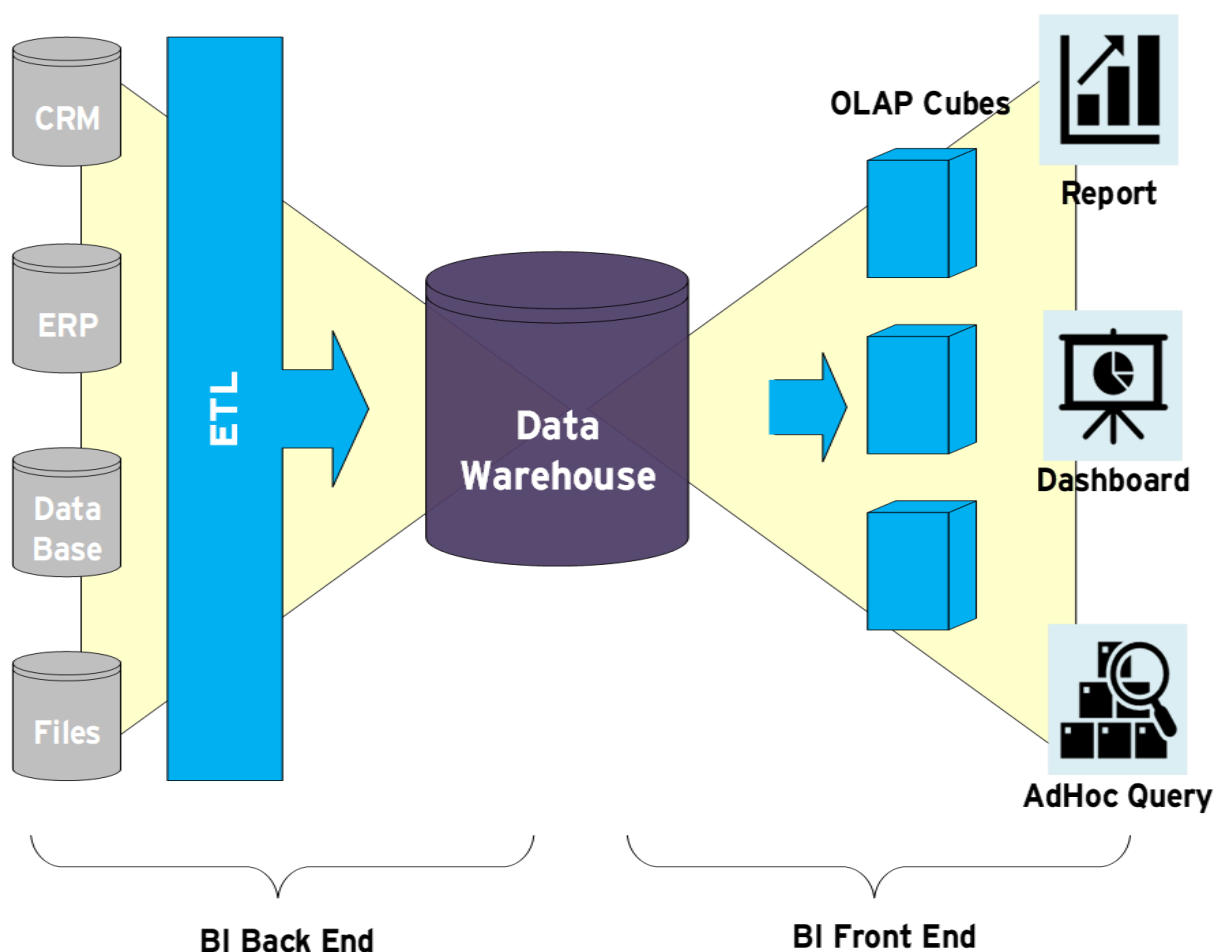


Figure 13: Data warehouse solution

3.7. Integration Architecture

The bidder may consider a government service bus (GSB) along with API management as an enterprise integration architecture model based on Service Oriented Architecture (SOA). ISTD expects the System will require the development of inbound and/or outbound interfaces with external systems, both inside and outside of the ISTD (e.g., interfaces may be required with external ministry systems, external service provider systems, etc.). The key capabilities to be considered under proposed solution for GSB and API management should include:

- repository
- publishing
- metering
- version manager
- analytics
- throttling
- Central management portal

The bidder should describe the approach to designing, documenting, developing the integration layer, and testing interfaces and integration points that are required for the implementation of the proposed System. The bidder should propose a decommissioning strategy in alignment with its interface/integration approach and provide for the expected use of "throw-away" interfaces in the event that legacy systems remain operational concurrently, until deemed ready to be decommissioned.

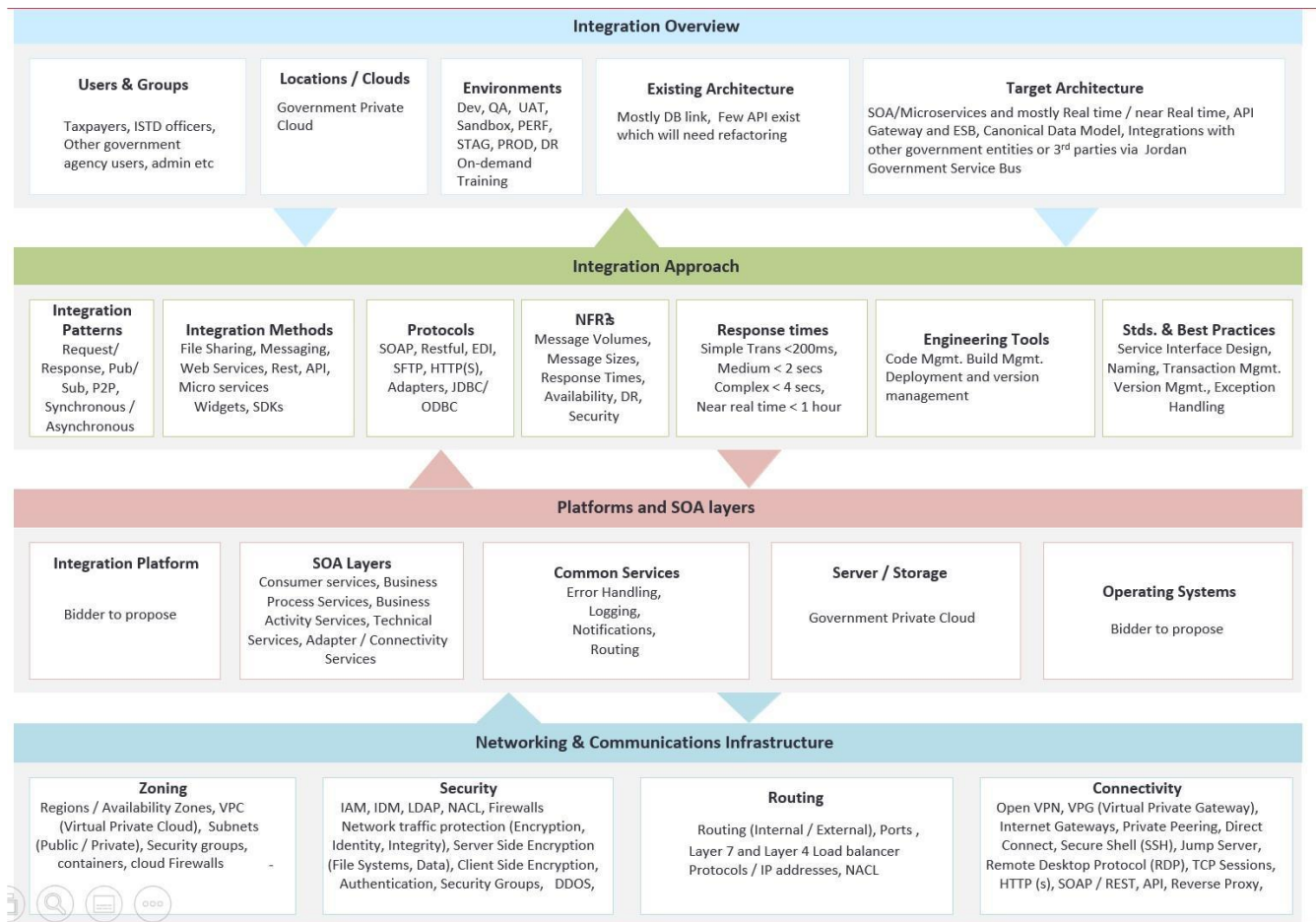


Figure 15: ITAS Integration Framework

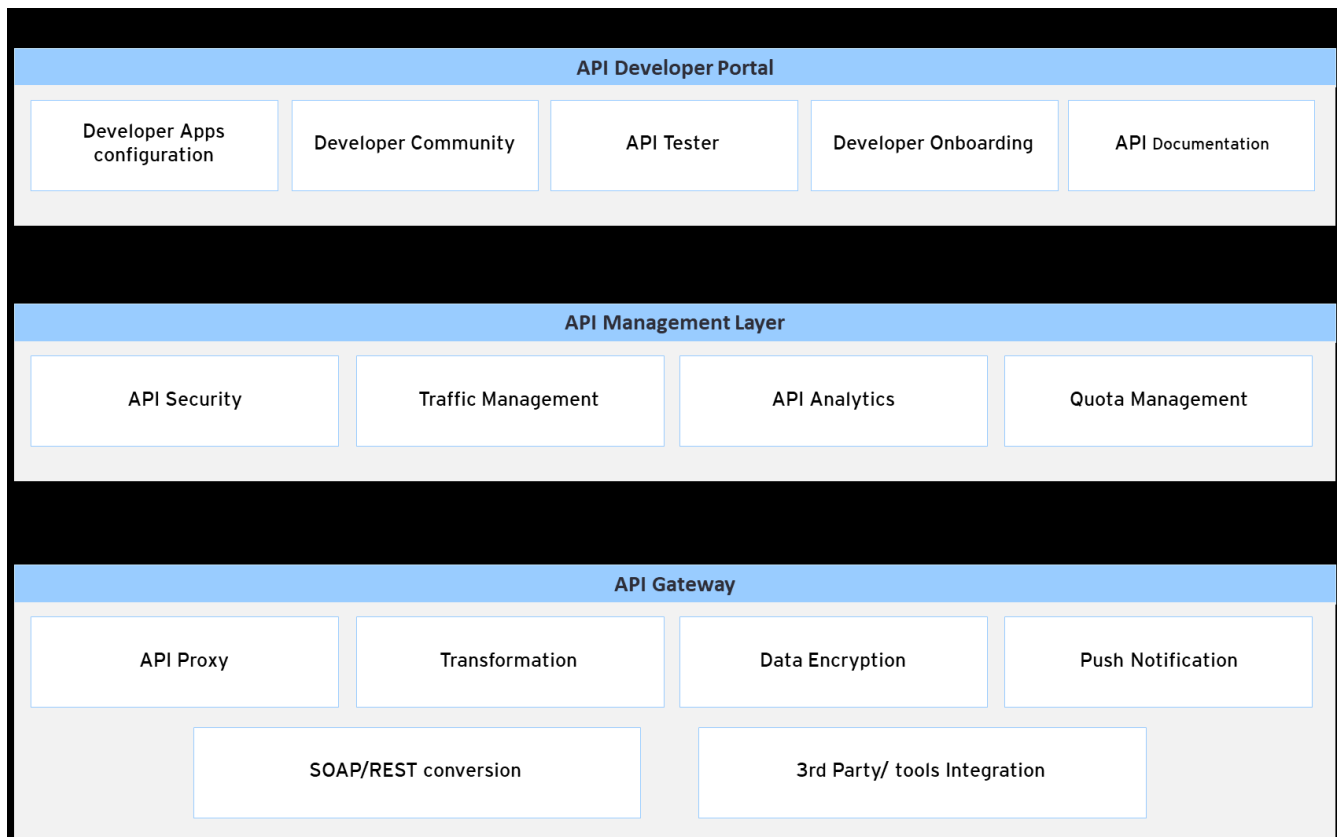


Figure 16: ITAS API Management Framework

The Bidder is required to undertake the following broad level activities under the integration layer:

- Integration layer design, development, test, and implementation
- The Bidder is required to design the integration layer components to facilitate exchange of information between ITAS and stakeholder systems
- The exchange is to be based on APIs
- The Bidder is expected to undertake a detailed study of the information exchange between ITAS and other stakeholder systems to identify the list of APIs to be implemented.
- Authentication and authorization
- Bidder will set up the requisite process as well as system to build, operate & manage and sustain APIs for ITAS in a secured and controlled environment.
- Authorized stakeholder systems approved by ISTD authorities shall be able to communicate with the APIs
- Bidder should create an administrative portal to enable stakeholders to have a user account to manage their services
- License Key shall allow enabling of various services for a given stakeholder
- The admin portal shall allow the port or the authorized agency to manage the license keys for the stakeholders
- The Bidder will design authentication services for stakeholders:
- Test APIs in sandbox
- Configure authorization policy for new APIs
- Grant users access to the available APIs and associated properties
- Blacklist / suspend / block access
- API Definition

- The Bidder is required to study the information exchange between ITAS and other stakeholders to design the system requirements for the API layer
- Based on the solution description and requirements, the Bidder is required to develop standardized APIs to expose ITAS services to stakeholders as well as consume stakeholder APIs required for ITAS
- Bidder will define API specifications document which would contain information including description, author/owner, input, output parameters
- A test environment should be made available which should have similar catalog to the production environment with API stubs for testing purposes before migration to production environment.
- API publishing
- The Bidder should define and establish the procedure to publish new APIs in test and production environment in discussion with the ISTD authorities
- The solution proposed should have a catalog to register the APIs which are being published in production environment
- The Bidder will be required to support stakeholders in functional, security and performance testing prior to publishing APIs in production
- Version Control
- Bidder is expected to establish procedure for managing version control of the APIs up to retire / archival of APIs
- The proposed solution should support in API version control
- The system should support in versioning and roll back as per requirement
- The proposed system should support retiring and archival of APIs as a part of the API lifecycle management
- API Governance
- The API lifecycle should be defined by the Bidder and corresponding governance procedures should be established after discussions with the port authorities
- The Bidder should establish plan to define and enforce SLAs for consuming resources
- The proposed solution should support API throttling wherein traffic is throttled or a notification may be issued to the appropriate authority / owner
- The system should generate reports on SLA and allow configuration of SLA parameters
- The proposed solution should allow API metering
- The Bidder is required to configure the system to regulate the recording and limit the usage of APIs for stakeholders as per the policy controls
- API security
- The proposed framework developed by the Bidder should define the API security guidelines
- The proposed system should inspect the headers for APIs genuineness before acceptance.
- It should apply all security checks e.g., DDoS Attacks, XML Denial of Service (xDoS), Slow down or disable an XML based System, Message Snooping, XML Document Size Attacks, XML Document Width Attacks, XML Document Depth Attacks, Jumbo Payloads, Recursive Elements, Public Key DoS, XML Flood, Resource Hijack etc. to ensure rightful and secured access to API consumers
- The MSP is required to configure the proposed solution to meet the security requirements and support in undertaking security review of the API before being published to the production environment
- Audit Support
- The Bidder is required to configure the solution to allow audit logging of transactions.
- The audit logs should capture at least the following information:
- Credentials
- Transactional logs
- IP address and date / time stamp
- Session details (time, id, key)

- License key
- Reporting
- The Bidder is required to configure the solution to generate reports on API usage and performance
- The solution should be capable of generating reports/online view that shows API related information such as
 - most/least accessed APIs
 - highest/lowest/average response time
 - data volumes transferred
 - transfer rates,
 - maximum and minimum response time over a period of time
 - rejected API requests
- API catalogs
- API versions
- List of stakeholder wise registered APIs
- API throttling limits (hard and soft)
- API notifications
- API activity – port wise
- API activity customer / partner / vendor wise
- Blocked / Hot listed / Archived / Suspended APIs

3.8. Interfacing Systems

The winning bidder shall integrate the ITAS with GSB through supporting APIs and message communication using REST APIs.

Integration with stakeholders could be implemented using SOA architecture, Microservices architecture (MSA), or hybrid solution. It is the winning bidder responsibility to build or update Web Services or APIs to integrate with Stakeholders.

A Government Service Bus (GSB) is a pattern of middleware and software infrastructure that enables Service Oriented Architecture (SOA) by acting as middleware through which a set of reusable Government Services are made widely available. It unifies and connects services and applications within the government of Jordan and provides a framework within which the capabilities of business' applications are made available for reuse by other applications throughout the organization and beyond. It also provides a messaging framework through which government e-services exchange information.

IBM API Connect is an end-to-end solution that allows users to create, secure, manage, socialize, monetize and analyze APIs. It provides a powerful set of capabilities from turning backend RESTFUL services into managed services. This is done by publishing APIs to API Gateways, while enforcing lifecycle and governance controls on those APIs. API Connect enables users to expose APIs, through a developer portal, targeting application developers both inside and outside their organization. Additionally, the solution's analytics tooling helps API providers and API consumers had better understand the health and consumption of deployed APIs.

Note: Integration with stakeholders has to be system-to-system integration by implementing web services through GSB platform. However, if there are limitations for GSB integration for specific Stakeholder(s), integration will be done as user role integration through screens to be accessed by stakeholder(s) or attaching specific files or documents to the application according to MoDEE team approval.

The Bidder is required to undertake the following broad level activities under the integration layer:

- ▶ Integration layer design, development, test, and implementation
- ▶ The Bidder is required to design the integration layer components to facilitate exchange of information between ITAS and stakeholder systems through GSB
- ▶ The exchange is to be based on Restful APIs
- ▶ The Restful APIs should comply with GSB requirements and should published there on both staging and production environments
- ▶ The Bidder is expected to undertake a detailed study of the information exchange between ITAS and other stakeholder systems to identify the list of APIs to be implemented.
- ▶ Authentication and authorization based on GSB requirements
- ▶ Bidder will set up the requisite process as well as system to build, operate & manage and sustain APIs for ITAS in a secured and controlled environment.
- ▶ Authorized stakeholder systems approved by MoDEE/ISTD authorities shall be able to communicate and consume the APIs
- ▶ API definition, documentation and usage
- ▶ The Bidder is required to study the information exchange between ITAS and other stakeholders to design the system requirements for the API layer
- ▶ Based on the solution description and requirements, the Bidder is required to develop standardized APIs to expose ITAS services to stakeholders as well as consume stakeholder APIs required for ITAS
- ▶ Bidder will define API specifications document which would contain information including description, author/owner, input, output parameters
- ▶ A test environment should be made available which should have similar catalog to the production environment with API stubs for testing purposes before migration to production environment.
- ▶ The Bidder will be required to support stakeholders in functional, security and performance testing prior to publishing APIs in production

Note:

- ▶ The existing integrations with ISTD's e-Services are explained in the ITAS Annexure C- ISTD Interfacing Systems
- ▶ A detailed list of the ITAS integration touch points is provided in the ITAS Annexure B - ITAS Technical Requirements.
- ▶ Any additional integration points that may come up during requirement gathering phase not mentioned against the associated ministries/3rd parties listed in the RFP, will be the winning bidder responsibility to build these integration points. Corresponding additional effort (man-hours) as well as fee to be approved by ISTD based on the rate card provided by the winning bidder in the commercial proposal.
- ▶ Any process or subservice related to the ITAS and essential to complete end-to end services, will be winning bidder responsibility to build these process/subservice without any cost.

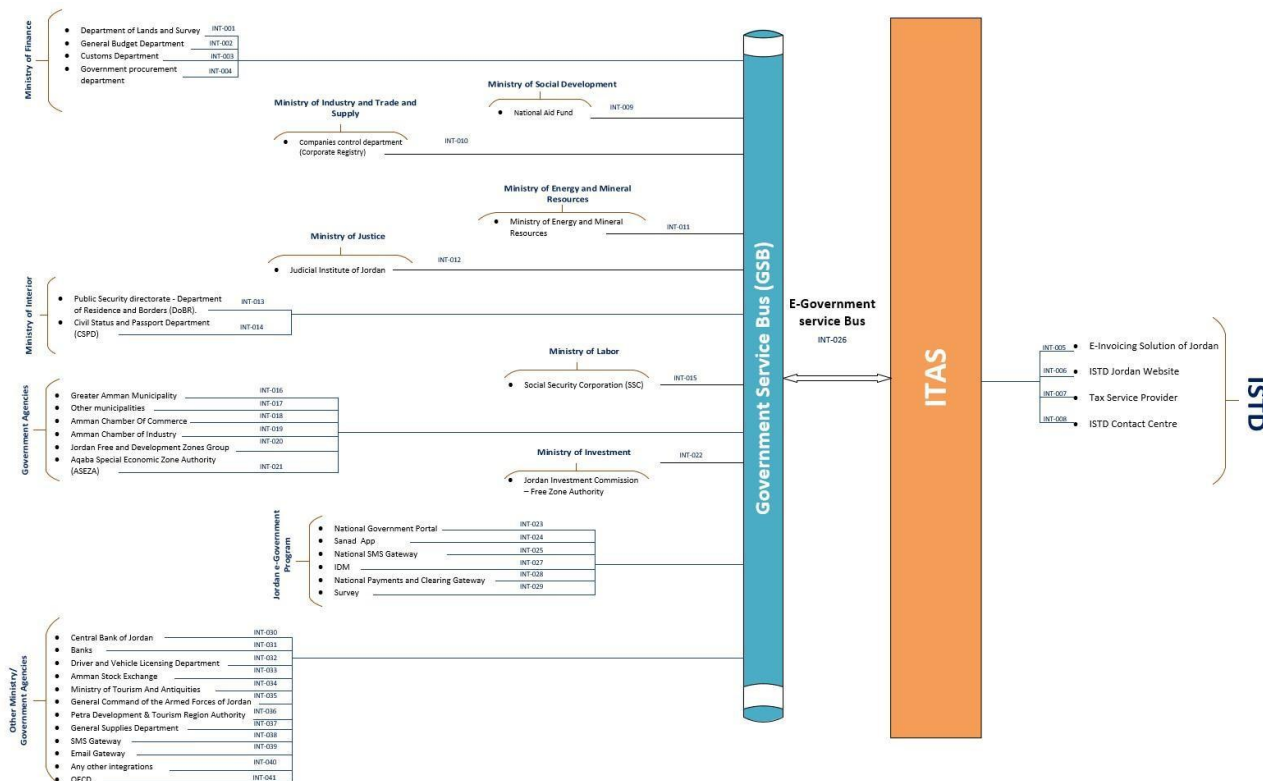


Figure 14: ITAS Interfacing systems

3.9. ITAS portal and CMS

The ITAS portal will act as the primary channel for various stakeholders to interact with ISTD. Various users like the Taxpayers, Tax agents, Legal Representative (refer "Functional requirements section") must be able to access the ITAS through a single unified Taxpayer Services portal. The user must be allowed to access their personalized Home page through their user credentials after being validated through a strong firewall. The portal should provide self-service support to the end users to manage their user profile information apart from enable them in performing their taxpayer functions such as submission, updating, review and track the real time status of various requests for registration, filing returns, payment of Tax and Penalties, request refunds, initiate objections etc. There should be single point for the users to receive all the Notifications, Correspondences and alerts based on predefined triggers.

The portal should be bilingual to support users to interact in both English and Arabic and should allow the user to toggle between the two language options. The Portal should be responsive to be accessed from multiple devices or varying sizes and resolutions through all the leading browsers and operating systems. The ITAS should record the user and transactions logs to ensure proper audit trail. The ITAS portal should be integrated with Jordan's National e-Government portal.

3.10. Mobile app

The ITAS mobile application(s) (with native support for iOS and Android) will provide general information and access to all ITAS services for the end users (taxpayer and ISTD tax officer apps).

The Winning bidder should define, publish and expose APIs from ITAS which can be consumed by the SANAD App for the Taxpayers to perform selected Tax functions.

3.11. Implementation Requirements

Please refer to Section 4 and Section 5 for details.

3.12. Business Process Management

Workflow management is the integral component of ITAS that not only facilitates workflow but also supports other functions such as document and content management. Some of the features of the workflow management tool are given below.

- ▶ Workflow engine supports work management at the operational level, Self-Assessment of Taxes including creating activities, assigning work, reporting, escalation, and administration of workflow.
- ▶ Business Rules Engine is a centralized business rule repository for storage and management of business rules and data validation rules.
- ▶ Document management provides the ability to store and retrieve electronic versions of documents and manage the retention and destruction of electronic records in accordance with legislative requirements.
- ▶ Content management is the enterprise functionality for initiating, approving, storing, maintaining, and publishing approved content for informational, transactional, interpretative, and correspondence products, including marketing and education materials.
- ▶ Case management manages all types of workflow for an individual case, e.g., a taxpayer audit, a taxpayer debt, a disputed assessment, an outstanding return, and responding to taxpayer correspondence.
- ▶ Generate determined management information for all case types, such as cases created, cases closed, cases still outstanding, elapsed time for actions, and maintain an audit trail of any accesses or adjustments made.

3.13. User Experience

The ISTD expects the bidder to understand the current challenges, future requirements, and journeys of ISTD users (taxpayers, tax officers and others) and accordingly address the UX requirements in the ITAS design as per the Jordan Government Websites Standards (GWS). Refer Annexures (H, I, J)

The Jordan Government Websites Standards (GWS) make the Jordanian government websites more usable, user-centric, and universally accessible. Additionally, it brings uniformity in the quality of content and provided services. Compliance to these guidelines will ensure a high level of consistency and uniformity in the content coverage and presentation that will lead to further excellence promotion in the Jordanian Government web space, whereby all government entities are strongly encouraged to comply with these guidelines

3.14. Training

The ISTD expects that the bidder should provide comprehensive initial and ongoing training over the life of the System. Describe the approach to training and knowledge transfer. The response should include the following:

- ▶ An overall training and knowledge transfer strategy
- ▶ A training needs analysis that documents all skills gaps by role and the number of training sessions that would be required to be conducted for different stakeholder groups
- ▶ A training plan that specifies the required courses/curriculum by role and how the training should be delivered
- ▶ A knowledge transfer plan that specifies the required courses/curriculum by role, knowledge achievement milestones, and a description of how the knowledge transfer should be conducted.

A plan for development and delivery of training and knowledge transfer modules, including on-site instructor-led training in Arabic.. The on-site instructor-led training should be held in ISTD headquarters in Amman.

The winning bidder is required to prepare, present and execute plan of knowledge transfer and training for identified stakeholders among different users' types. It should ensure that all types of users and administrators are prepared to operate ITAS and to take over future enhancements or support the ITAS applications (especially in technical area).

The winning bidder to provide financials (in Fin-Form 4.6) for both the options, stated below:

- Option 1: To train 15% of the concerned team members (around 1350+ personnel) as Training the Trainer.
- Option 2: To provide a training plan to train all concerned team members (around 1350+).

Winning bidder must price comprehensive, professional-standard training programs covering all technologies included in the proposed solution. These training tracks must follow the official guidelines and certification standards of the Original Equipment Manufacturer (OEM) or the respective technology owner. Each track should represent a full training path leading to recognized certification, not just introductory sessions.

4. Scope of Work

This RFP is published for selection of bidder to implement the ITAS. The scope of work as part of this engagement comprises supply of ISTD application modules, installation of associated peripherals and accessories, Design, Development, Implementation, Data Migration, Operations, and Maintenance of ITAS software system.

4.1. Overview

The scope of work is to implement the Integrated Tax Administration System (ITAS) to ensure a seamless experience and enhanced services for taxpayers in the Hashemite Kingdom of Jordan. The scope consists of system/solution installation, design, customization, implementation, data migration and the operation and maintenance of the system. All the deliverables such as BRDs need to be in Arabic as well as in English while few technical deliverables such as High Level Design (HLD), Low Level Design (LLD) etc. could be in English.

The operation and maintenance phase will be for a period of ten (10) years after Preliminary Acceptance of the system including helpdesk support for ISTD stakeholders. Following the completion of the 10-year support period, the contract can be further extended at the discretion of the ISTD.

4.1.1. Approach & Methodology

ISTD intends to implement ITAS within 2 years from the selection of the technology vendor. Accordingly, the bidder is expected to provide an implementation approach considering all the functional and technical requirements for ITAS.

Based on ISTD's objectives, recommended implementation approach and requirements are given below. However, the bidder is expected to come up with a detailed implementation approach and plan to meet the below requirements.

- The new ITAS system will be implemented at the same time and expected to take two years to implement.
- Design considerations such as use of common data structures across ITAS.
- Data migration from existing core tax system to ITAS would be a critical activity and bidder is expected to conduct a deep dive study into existing data setup to come up with a data migration strategy.
- Communication must be in Arabic and if there is a foreign expert, the translation of Arabic is the responsibility of the winning bidder.

4.2. Development

The Implementation of the ITAS solution will have the below stages of a software development cycle.

4.2.1. Requirement Gathering

The winning bidder must perform a detailed assessment of the business, functional, nonfunctional, and technical solution requirements as mentioned in this RFP. The winning bidder is expected to conduct out a detailed requirement-gathering workshop in coordination with the ISTD for understanding the requirements. During this stage, the winning bidder should perform a list of activities as below but not limited to :

- ▶ Develop and follow standardized templates for capturing requirements and system documentation
- ▶ Conduct workshops, discussions, meetings, etc. to capture requirements and to review the same
- ▶ Prepare, Amend, refine, and enhance the Business Requirements Document (BRD)
- ▶ Prepare a Software Requirement Specification (SRS) using the BRD
- ▶ Maintain a Requirement traceability matrix (RTM) from BRD stage onwards for the entire implementation.
- ▶ Winning bidder is required to update the BRD, SRS and RTM as and when any enhancements / modifications are made to the overall solution to ensure that the documentation is always updated for the entire duration of the Contract.
- ▶ Obtain the required signoffs from ISTD and MoDEE.

The BRD and SRS document together should necessarily contain the following details:

- ▶ User groups, roles, and types of access
- ▶ System attributes
- ▶ Use case diagrams for business process
- ▶ Activity diagrams for business process identified
- ▶ Business rules, parameterization requirements
- ▶ Sequence diagrams for business processes identified
- ▶ Functional logic and checks expected from the system while execution of the project
- ▶ Software components which will be used including ITAS and other operations systems, ERP, electronic workflows, etc.
- ▶ Reporting and analytics requirements
- ▶ Mobile requirements
- ▶ Interfaces with other ISTD systems and external systems
- ▶ Security requirements
- ▶ Audit and application logging requirements
- ▶ Archival requirements
- ▶ Migration requirements
- ▶ SLA monitoring requirements
- ▶ Functional, non-functional, performance and external interface requirements

- ▶ Other details which are required to clearly articulate software requirements

While business requirements of the proposed ITAS are attached to this RFP some of the requirements, processes, specification etc. may undergo changes in this stage. The winning bidder needs to consider this fact while submitting its proposal. Such changes will not be considered as a Change request till the SRS approval stage. All major changes post this for each phase shall be handled through change control process

4.2.2. Design

The winning bidder should design business applications of ITAS in two levels i.e., High-Level Design and LowLevel Design. The Winning bidder shall ensure that the design complies with the requirements outlined in this

RFP

In the high-level design, the winning bidder should develop an overview of proposed ITAS highlighting the main components and interfaces. A high-level design shall provide an overview of a system, product, service, and process. This will be useful to identify critical components and interactions of all components. The design will necessarily cover the following:

- ▶ Solution architecture covering minimum technical architecture view, data architecture view, application architecture view, network architecture view, data center architecture view, security architecture view, end user computing view
- ▶ Software components and their interaction with each other
- ▶ Customizations which are being made in the products being installed
- ▶ Interfaces with other systems internal and external
- ▶ Integration architecture
- ▶ Data architecture
- ▶ Format of all input screens including data entry requirements
- ▶ Documentation of detailed data requirements (input /output /outcome) defined during SRS
- ▶ Flow of screens and controls of the software
- ▶ Format of all reports that would be generated by the proposed system
- ▶ Access control mechanisms, data security and audit trails to ensure that databases are not tampered with or modified by unauthorized users
- ▶ User input forms and screen designs for all processes
- ▶ Report design for operations users and end users
- ▶ Dashboard design for leadership and senior stakeholders
- ▶ Enablement of Single sign-on, access control measures
- ▶ Other software components as required

In the low-level design, winning bidder shall refine the components defined in the High-Level Design for the purpose of actual software development. During this stage the winning bidder is expected to, but not limited

to, deliver the following documents:

- ▶ Detailed Application, Data, and Infrastructure Architecture Definition Document
- ▶ A Component-Level Design Specification document including the user interface prototypes, UML use cases, UML designs, Conceptual Data Model, Logical and Physical Data Model for different application modules, API Specifications (Signature /Contract), Stored Procedures Specifications etc.
- ▶ Low Level Design, with detailed algorithms and logic of the important API's stored procedures etc.
- ▶ Requirement Traceability matrix with updated design features against the requirements

4.2.3. Development and Customization

In this stage, the Winning bidder shall develop the required business applications of ITAS as per the gathered requirements and design. The winning bidder must try to use the open-source technologies and products are utilized for solution platform. In this phase of development, the winning bidder's development team would construct different modules of the applications using the specified tools. Development shall happen in parallel for different modules and wherever there is an integration need, manual stubs would be utilized so that there is no dependency on any other module and construction can happen seamlessly. During development of the business applications of ITAS, the winning bidder must note the following:

- ▶ Development and Testing Environment: The winning bidder should host the development and testing environment on an infrastructure in its own data center along with all requisite tools and licenses.
- ▶ Processes: While developing the application and carrying out on-going maintenance, the processes may be in compliance with a mature CMMI level. The source code documentation should be done in a descriptive and intuitive manner in an easy-to-understand language.
- ▶ Intellectual Property Rights (IPR): The ownership and Intellectual Property Rights (IPR) of the source code of the developed solution would be in the name of the ISTD. In case of a COTS product, the IPR of any customization done on the COTS product would be in the name of Jordan Government – ISTD.
- ▶ User Friendly: The application software developed by the winning bidder must be user friendly so that users can access it without having extensive training.
- ▶ Parallel Activities: The lifecycle for each business application should be independent, i.e., multiple teams should work in parallel to complete the implementation as per the project timelines defined in this RFP.

The winning bidder should procure any other tools and accessories required to complete the integrated solution per requirements as part of the solution. With respect to applications, the winning bidder should ensure that the following is adhered to:

The winning bidder will be responsible for supplying the application and all other related licenses of related software products, solutions or components which are required to meet the functional and technical requirements in this RFP including database licenses, in the name of ISTD, and installing the same so as to meet ITAS requirements. Operating system licenses and underlying infrastructure shall be provided by MoDEE/ISTD.

The winning bidder will be responsible for supplying other tools, accessories, documentation required to make the integrated solution complete as per the requirements. Tools and accessories shall be part of the solution.

During this stage the winning bidder is expected to, but not limited to, deliver the following:

- ▶ Updated System Requirement Specifications (SRS)
- ▶ Technical and product related manuals

- ▶ Installation guides
- ▶ Data Model Descriptions
- ▶ Frequently asked questions guide
- ▶ User manuals and Training Material
- ▶ Change management histories
- ▶ Version control mechanism and data
- ▶ SOPs, procedures, policies, processes, etc.
- ▶ Developed Source code of the applications with older versions, if any

The above documents need to be updated after each phase of project and to be maintained/periodically (at least once in a year) updated during entire project duration. All documents and source code of the ITAS system will be the property of MoDEE/ISTD.

4.2.4. Quality Management (Testing)

In order to provide Quality Management, the winning bidder is required to perform the activities mentioned below, noting that any additional related activities needed for the proper functioning of the system should be provided by the winning bidder and its cost should be included in the fixed lump sum price submitted by the winning bidder:

- ▶ Perform agile testing as it will be an integral part of the software development, where the whole development team will be conducting the testing on the developed features and functionalities and check behavior of the outcomes according to the expectations and requirements of MODEE/ISTD team:
- ▶ Conduct sprint units testing for eservices and integrations points.
- ▶ Conduct sprint test.
- ▶ Assign a dedicated Quality team to ensure quality of project deliverables or software through the related set of (Verification and Validation) activities.
- ▶ The winning bidder Quality team shall be responsible for performing all testing activities according to plans and procedures defined within the quality plan, and as per the requirements stipulated in this document
- ▶ Prepare a detailed Quality plan scope that should include all project phases, deliverables, and artefacts of any type relevant to the project nature like Portals, websites, e-Services software, documentation, etc. The winning bidder Quality team shall be responsible for performing all testing activities according to plans and procedures defined within the quality plan, and as per the requirements stipulated within this document.
- ▶ Provide all Quality deliverables which ensure that all related activities are done successfully. This includes but not limited to Test Plans, Test Case Scenarios including acceptance test scenarios, Testing results/reports, Testing Summary report, Defect (Bug) report and other required/proposed artefacts.
- Prepare the testing/staging environment to be identical to production environment in the following points:
 - Testing environment is fully Integrated to all web services and web forms

- Testing environment is fully integrated to staging e-payment gateways and shared government services
- ▶ Ensure proper deployment from staging environment to the ultimate Production environment after getting the approval from MODEE in corporation with ISTD. These environments are to be prepared and set by the winning bidder on the GPC.
- ▶ Perform all needed activities in the User Acceptance Testing that should be done in cooperation with MoDEE\ Third party and ISTD, all bugs and defects should be solved in order to get the approval on ITAS launching.

NOTE: MODEE reserves the right to perform their own functional and non-functional test including security, performance, quality and customer journey test on the solution (2 rounds test) and provide the reports to the winning bidder to apply bug fixing and recommendations to ensure system functionalities this will be done in each phase.

- **Prior conducting performance test, winning bidder, with collaboration with MoDEE project manager, must deliver the requirements listed in Annexure K - Performance Test Checklist.**
- In case an additional round of testing is needed after the official 2 rounds, the cost will be covered by the winning bidder.

Following are the estimated cost in JD for each testing type for each round:

#	Item Description	Additional round (JD)
1.	Quality Test	620 (per service)
2.	Customer Journey Test	440
3.	Performance	125
4.	Security	730

The table below depicts the details for the various kinds of testing activities required for each phase of the project:

Table 21: Quality and Management (Testing)

SL No	Type of Testing	Owner	Scope of work
1	Testing Documentation	Winning Bidder	The winning bidder should create a test strategy document and based on the test strategy a test plan needs to be created For performance testing, a separate performance strategy document needs to be created Based on the business processes & use cases, test scenarios & scripts need to be defined and mapped to the expected test result Requirement Traceability Matrix needs to be updated based on the test scenarios and test cases. All documents will be reviewed and approved by the ISTD\MoDEE
2	System Unit Testing	Winning Bidder	Perform component testing in environment where the development work is being carried out, hosted in the bidder's data center Share the unit test results with the ISTD\MoDEE Should be performed through manual and/or automated methods Automation testing tools may be suggested by the winning bidder
3	System Integration Testing	Winning Bidder	Perform System integration testing in an environment hosted in bidder's own data center Prepare and share with ISTD\MoDEE the integration test plans and test cases Perform integration testing as per the approved plan Integration testing to be performed through manual and / or automated methods

4	Performance Load Testing	and Winning Bidder	
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SL No	Type of Testing	Owner	Scope of work
			The system should meet efficiency targets to serve volumes of transactions and number of users. Please find the performance measures defined in annex K. The winning bidder shall propose the values of quality metrics mentioned in annex K and they should adhere to industry best practices which are subject to approval, rejection or modification by MODEE in cooperation with ISTD. Winning bidder, in collaboration with MoDEE project manager, must assure the checklist listed in Annex K is ready prior conducting the performance

5	Security Testing (Penetration and Vulnerability Testing)	Winning Bidder ISTD	The solution should demonstrate compliance with security requirements as mentioned in the RFP including but not limited to security controls mentioned in the The solution shall pass vulnerability and penetration testing for rollout The solution should pass web application security testing for the ITAS The Winning bidder should carry out security and vulnerability testing on the developed solution. Security testing to be carried out in the exact same environment/architecture that would be set up for production. Security test report and test cases should be shared with the ISTD\MoDEE Testing tools to be proposed, procured, and provided by the Winning bidder
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	Type of Testing	Owner	Scope of work
			During O&M phase, penetration testing to be conducted on yearly basis and vulnerability assessment to be conducted on half-yearly basis ISTD to provide testing acceptance/signoff

6	User Acceptance Testing	ISTD	1. ISTD and MoDEE will perform User Acceptance Testing or MoDEE may choose to perform the same by contracting an associated MoDEE vendor. 2. The Winning bidder will need to prepare the User Acceptance Testing test cases and execution strategy 3. UAT will have to be carried out in the exact same environment/architecture as the one set up for Production 4. The Winning bidder should fix bugs and issues raised during UAT and seek approval on the fixes from ISTD\MoDEE 5. Changes in the application as an outcome of UAT shall not be considered as a Change Request. The Winning bidder will need to rectify the observations raised.
7	Beta Testing of ITAS	ISTD	Beta testing before the Go Live shall be conducted by the ISTD to understand the end user experience The Winning bidder to prepare Beta acceptance testing test scenarios and test cases Beta testing to be carried out in the exact same environment/architecture that would be set up for production The Winning bidder should fix bugs and issues raised during Beta testing and get approval on the fixes from the ISTD before Go Live Changes in the application as an outcome of Beta testing shall not be considered as Change Request. The Winning bidder must rectify the observations The MODEE/ISTD to provide final Beta signoff
8	Application Benchmarking	Winning Bidder	The Winning bidder needs to perform a benchmarking of the application software to meet the scalability requirements provided below: The ITAS should be benchmarked to 10x of the API transaction volumes (for identified transactional API e.g., upload monthly return API, etc.) within 6
SL No	Type of Testing	Owner	Scope of work

			months post Go-Live. Simulated data covering real life scenarios should be used for all benchmarks. Benchmark could be done in GPC managed by MoDEE. Production data should not be used for testing The benchmarking of the application software shall be done by a third party engaged by the Winning bidder. The Winning bidder will bear the costs of such benchmarking exercises. The third party will certify the system's ability to handle the requisite load. In case the system fails to handle the expected loads, the AP will have to take corrective action Inputs from the MoDEE/ISTD will be incorporated by the Winning bidder before performing the benchmarking exercise
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Note:

The winning bidder is required to provide the following information in the technical proposal in relation to the Quality Management and validation and demonstrate the approach and components through which the quality

plan shall be implemented. The proposal should provide adequate explanation regarding the proposed Quality management plan, including but not limited to:

- ▶ Describe methodology and quality standards for the overall Quality Management and bidder's professional qualifications (like Quality certificates/accreditation) in quality management.
- ▶ Identify and describe the process for reviewing the test plans, test cases, and test results, identify the defect tracking processes, test environments, test roles and responsibilities, and test phase entrance/exit criteria.
- ▶ Comply that testing/staging environment will be identical to production environment as described above.
- ▶ Describe the project's quality practices, including but not limited to:
 - ▶ The set of reviews and checkpoints for the project, including entry and/or exit criteria; hold those reviews, and measure against entry/exit criteria.
 - ▶ The standards and KPI's to be used to measure project deliverable quality.
 - ▶ The Quality metrics to be used to measure project deliverable quality.
- ▶ Identify and describe the testing tools should be used by the winning bidder to perform all required testing types to measure of project deliverables quality and final products.
- ▶ Provide a list of deliverables for the Quality Planning and Management, as mentioned in the deliverable section below, and as per the winning bidder proposed approach.
- ▶ The winning bidder needs to provide the details of the testing strategy and approach including details of intended tools/environment to be used by the AP for testing in its technical proposal.
- ▶ The winning bidder shall work in a manner to satisfy all the testing requirements and adhere to the testing strategy outlined. The winning bidder must ensure deployment of necessary resources and tools during the testing phases. The winning bidder shall perform the testing of the solution based on the approved test plan, document the results, and shall fix the bugs found during the testing. It is the responsibility of the winning bidder to ensure that the product delivered by the winning bidder meets all the requirements specified in the RFP. The winning bidder shall take remedial action based on outcome of the tests.
- ▶ The winning bidder shall arrange for environments and tools for testing as envisaged. Post Go-Live; the production environment should not be used for testing and training purpose. If any production data is used for testing, it should be masked and it should be protected. Detailed process in this regard including security requirement should be provided by the winning bidder in its technical proposal. The process will be finalized with the Winning bidder.
- ▶ The cost of rectification of non-compliances shall be borne by the winning bidder.

The winning bidder is required to provide the deliverables mentioned below, noting that any other related deliverables needed for the proper functioning of the system should be provided by the winning bidder and its cost should be included in the fixed lump sum price submitted by the bidder. Quality management documentation that will cover the different knowledge areas, including but not limited to:

- ▶ Release Quality plan including and not limited to:

- ▶ Criteria for starting and ending each testing stage (to be determined by MODEE quality team) Entry and exist criteria for each testing stage.
- ▶ Schedules for third party UAT sessions for 2 rounds (Smoke test, Quality functional Test, Customer Journey tests as per standards highlighted in Annexure J – MODEE Customer Journey Standards Compliance Report 2020, Performance test, & Security test).
- ▶ Duration for each test will be determined according to e-service complexity and the project plan.
- ▶ Schedule for Business owner UAT session.
- ▶ Schedule for Fixing defects
- ▶ Duration of fixing defect will be determined in cooperation with MODEE project manager.
- ▶ Risk evaluations.
- ▶ UAT Plan document.
- ▶ Quality Roles and responsibilities
- ▶ Detailed, searchable, cleared test cases and test scenarios with representative data set for each eservice document (positive and negative scenarios).
- ▶ Quality metrics and Key Performance Indicators document
- ▶ Performed UAT sessions comply with UAT certificate signed from the business owner.
- ▶ SANAD Mobile application if (if applicable)
- ▶ Test reports for each e-service include:
 - Functional end-to-end test report
 - Integration test report (for web services, web forums and backend systems)
 - Usability test report
 - e-payment integration test report
 - Compatibility test report (including all supported browsers, screens, devises, Operating systems)
 - "Consuming APIs" test report.

*Please refer ITAS Annexure B - ITAS Technical Requirements for detailed requirements.

4.3. ITAS Application Support

4.3.1. ITAS Services Portal Solution

A web portal shall be implemented as part of ITAS. The ITAS Web Portal would be available to all stakeholders (companies, taxpayers, taxpayer's representatives, internal users, and Administrators) to perform various functions under the VAT ecosystem. The taxpayers will be able to use applications such as pre-registration, registration, e-return filing, e-payment, etc. The internal users will be able to access applications admin functions, assessment functions, support functions, MIS reports, etc.

The Winning Bidder shall be responsible for the following:

- ▶ Gather requirements and design the ITAS Web Portal.
- ▶ Development of the ITAS Web Portal.
- ▶ Hosting and subsequent maintenance of the portal in accordance with the service levels.
- ▶ Implementation of a robust portal security solution and its continuous improvement on an ongoing basis.
- ▶ Develop and update the portal's content in consultation with MoDEE/ISTD.
- ▶ Implement a "Content Management" framework and solution to allow authorized users of MoDEE/ISTD to manage publication of new content on the portal. As part of this, the Winning Bidder shall provide training to 5 users identified by ISTD on the following:
 - ▶ Overview of the ITAS Portal Content Management for MoDEE/ISTD.
 - ▶ Portal operations, functionalities, and features.
 - ▶ Re-design/enhance the portals whenever required on ISTD's request. Some of the key requirements of such re-design/enhancement shall be:
 - ▶ Leverage technological advancements for portal applications as they emerge and implement the same.
 - ▶ Implement basic design principles in portal design including use of consistent, unified common themes including a consistent unique stylesheet including fonts, colors, etc. and implement consistent look and feel and navigation.
 - ▶ Provide universal accessibility: The portal shall be accessible to all irrespective of technology, platform, devices of any kind.
- ▶ The Winning Bidder is expected to position appropriate qualified and trained manpower for ITAS development followed by Operations & Maintenance (O&M). In line with the manpower requirements (Ref. Section 5) winning bidder is expected to provide a detailed manpower deployment plan for the entire project duration including ITAS development and O&M.
- ▶ The portals should support both English and Arabic languages.

4.3.2. Business Intelligence and Analytics

A comprehensive BI and Analytics module is envisaged as part of ITAS. The features of the same is provided in Business Intelligence & Analytics. The Winning Bidder's scope of work regarding this would include:

- ▶ Procuring, commissioning, configuration, implementation, integration, deployment, and maintenance of Data Warehouse and Business Intelligence Solution for ITAS.
- ▶ The mechanism would also allow for alerts, reminders, etc. to be sent through a unified dashboard.
- ▶ Winning Bidder shall carry out a detailed requirement phase to produce a detailed functional specifications and design specifications, including detailing the data warehouse and BI module to be developed, system architecture design, design principles/considerations, etc.
- ▶ The solution must have analytics and customized reporting. Reports should allow for exportable format such as pdf, excel etc.
- ▶ Winning Bidder should propose tools that allow customizable reports. The generation of the report shall not impair the System performance.
- ▶ ISTD shall prescribe reports to be developed which will be identified at requirements stage as well as operations phase.

4.3.3. ISTD Call Centre and IT Helpdesk services by winning bidder

4.3.3.1. IT Helpdesk

- ▶ ISTD will provide a dedicated call center to provide support on ITAS system related queries/ technical issues of taxpayers and tax officials. ISTD / ISTD nominated agency will set up the call center, man it with the requisite L1 resources, provide a toll-free number for the call center and will bear the operational cost of this number. The call center will provide support in both English and Arabic languages.
- ▶ Winning bidder will be responsible to train the ISTD manpower for the resolution of L1 system related queries of users. This training will be done on periodic basis as required and mutually agreed. Winning bidder shall also provide required training material to call center staff. All ITAS application related queries which are not resolved by L1 call center team of ISTD would be re-routed through CRM to the winning bidder's IT helpdesk team for resolution and status update. Winning bidder shall also prepare and follow a troubleshooting guide in consultation with ISTD.
- ▶ Winning bidder will be required to integrate its CRM solution with ISTD call center system but shall not be responsible to procure, supply and maintain the required IT/ Non-IT infrastructure (Hardware/ Software) to run the operations of the call center.

4.3.4. External Applications

The Winning Bidder is expected to ensure smooth integration of ITAS with other Government / third party / external applications as required.

- ▶ Government Applications – Integration of ITAS may be required with other Government / Ministry applications for exchange of financial data / taxpayer information. The Winning Bidder needs to provision for any such integration as part of the ITAS solution design.

- ▶ Other external applications / Tax systems – It is envisaged that ITAS should be able to integrate with any other Taxes and related systems, as and when they are implemented in Jordan. The Winning Bidder needs to provision for any such future integration as part of the ITAS solution design.

4.4. Application Enhancement and Maintenance

The Winning Bidder will be responsible for operations and management of ITAS for a period of 10 years post Go-Live. ITAS maintenance and management support will include, but not limited to, troubleshooting, and addressing functionality/availability and performance issues and implementing change requests etc. For application maintenance and management, the Winning Bidder shall provide a team of full-time resources as mentioned in Operations and management support shall be undertaken whenever any maintenance and enhancement activity need to be carried out as per the mutually agreed plan or change request. The Winning Bidder shall ensure compliance with SLAs as indicated in this RFP. Any upgrades/major changes to the software shall be planned by the Winning Bidder while ensuring that the SLA requirements are met at no additional cost to ISTD.

Key activities to be performed by the Winning Bidder are mentioned in subsequent sections of this RFP.

4.4.1. Annual Technical Support

The Winning Bidder shall be responsible for providing annual technology support for the COTS/OTS components of ITAS, if any, as supplied by respective OEMs during the entire operations and management phase. It is mandatory for the Winning Bidder to take enterprise level annual support over the entire contract duration, at minimum, for the software(s) as mentioned in the Bill of Material provided by the Winning Bidder.

4.4.2. Application Support

Winning Bidder shall provide continuous support during phased roll out of ITAS as well as during O&M phase. The Winning Bidder shall address all the issues/bugs/gaps in the functionalities of ITAS vis-à-vis the signed-off FRS and SRS at no additional cost during the operations and management phase.

Winning Bidder shall establish a suitable technical helpdesk function with dedicated manpower for smooth functioning of ITAS. Winning Bidder shall be responsible for L2 and L3 support of ITAS, while the L1 support would be managed by the Call Center agents who would act as the primary contact for all technical / application related issues reported by ISTD Users / other stakeholders. Winning Bidder's responsibility would include:

- ▶ Ensure all ITAS related technical issues are logged into the Call Center CRM as reported by the Users/stakeholders
- ▶ Provide proper training to the Call Center agents for resolving L1 queries / issues
- ▶ Ensure automatic re-routing of L2 and L3 issues through CRM to Winning Bidder's technical helpdesk support team
- ▶ Track satisfactory resolution of the issue and communication to the User who reported the issue
- ▶ All required SLAs are followed while resolution of the reported issues

Note: The process of escalation and level of reporting shall be finalized in consultation with ISTD.

4.4.3. Change Management and Version Control

All planned or emergency changes to any component of ITAS shall be carried out through the approved Change Control Management process. The Winning Bidder needs to follow all such processes (always based on industry standard framework). For any change, Winning Bidder shall ensure:

- ▶ Detailed impact analysis is conducted
- ▶ All change plans are backed by roll back plans
- ▶ Appropriate communication on change required has taken place
- ▶ Requisite approvals have been received
- ▶ Schedules have been adjusted to minimize impact on the Production environment
- ▶ All associated documentation is updated post stabilization of the implemented change
- ▶ Version control is maintained for all software changes

4.4.4. Issue Identification and Resolution

Winning Bidder shall ensure that all Application issues and bugs are proactively identified through comprehensive STLC, so that ITAS performance does not impact a wider range of users. All the issues / bugs need to be logged and tracked for proper resolution through a bug tracking tool. Winning Bidder shall submit monthly Issue Logs on problems identified and resolved along with recommended solutions, to ISTD.

4.4.5. Release Management

The development and deployment of ITAS is envisaged to be in 4 phases. Each phase is marked with specific version release of the ITAS. The Winning Bidder needs to ensure overall release management and version control of ITAS as per phased implementation plan defined in above sections of this RFP.

The Winning Bidder shall carry out following activities as part of the release management process:

- ▶ Plan releases as per the requirements of ITAS implementation plan
- ▶ Build release packages for the deployment for approved changes (one / many) into QA/SIT/Pre-prod or Staging/production

4.5. ITAS Hosting Requirements

It is for information of the bidder that the ITAS solution will be hosted on a Government Private Cloud (GPC) which would be provided and managed by MoDEE. The implementation and hosting of ITAS would be governed by the GPC Policy. For details on GPC Services refer Section 3.2 Table 17.

4.5.1. Role of GPC in Cloud Hosting

Under the policy, GPC is expected to perform the following responsibilities:

- ▶ Act as the interface between GPC and ISTD.
- ▶ Security clearance for ITAS application for hosting on Cloud. Winning Bidder needs to undertake all requisite changes as suggested by MoDEE to meet the security requirements before the application is hosted at GPC
- ▶ Provide cloud hosting compliance requirements to the Winning Bidder
- ▶ DC DR configuration at infrastructure level in Active-Active or Active-Passive mode depending upon the solution requirements of ITAS. Winning Bidder will be expected to configure the application layer.
- ▶ Meet the Cloud scalability requirements of ITAS application on advance intimation
- ▶ Support in meeting relevant SLAs for technical services based on the requirements for ITAS.
- ▶ Monitor and govern SLAs agreed with Government Private Cloud (GPC)
- ▶ Technical support to Winning Bidder for meeting the hosting requirements for ITAS applications.
- ▶ Provide coordinates of landing site to the Winning Bidder for networking requirements and connectivity between the landing site and office of ISTD
- ▶ Support in periodic application and data back up

Refer Annexure B – ITAS Technical Requirements for further details.

4.5.2. Role of Winning Bidder in Cloud Hosting

The Winning Bidder shall be responsible for deploying the entire ITAS solution on a Cloud. The Winning Bidder shall be provided a Government Private Cloud (GPC) by MoDEE.

All the requirements/scope of work mentioned in this section shall be the responsibility of the Winning Bidder.

- ▶ The Winning Bidder shall deploy the developed ITAS Solution on a “Cloud” of the GPC, complying to different versions of ITAS Application
- ▶ The Winning Bidder is required to prepare and submit along with their technical proposal, the details of methodologies and computations for sizing and capacity of storage, compute, backup, network, and security
- ▶ There should be sufficient capacity (compute, network and storage capacity offered) available for near real time provisioning (as per the SLA requirement of the Contract) during any unanticipated spikes in the user load.
- ▶ The Winning Bidder will be responsible for adequately sizing the necessary compute, memory, and storage required, building the redundancy into the architecture (including storage) and load balancing to meet the service levels mentioned in the RFP.
- ▶ While the initial sizing and provisioning of the underlying infrastructure (including the system software

and bandwidth) would be carried out for the first year; subsequently, it is expected that the Winning Bidder, based on the growth in the user load (peak and non-peak periods; year-on-year increase), will scale up or scale down the compute, memory, storage, and bandwidth requirements to support the scalability and performance requirements of the solution and meet the SLAs.

- ▶ Ensure redundancy at each level
- ▶ Winning Bidder shall provide interoperability support with regards to available APIs, data portability etc. for MoDEE/ISTD to utilize in case of:
 - ▶ Change of Cloud Service Provider,
 - ▶ Migration back to in-house infrastructure,
 - ▶ Burst to a different cloud service provider for a short duration, or
 - ▶ Availing backup or DR services from a different service provider
- ▶ Undertake migration of the VMs, data, content and any other assets to the new environment created by the MoDEE/ISTD or any Agency (on behalf of the Government) on alternate cloud service provider's offerings to enable successful deployment and running of ITAS on the new infrastructure.
- ▶ MoDEE/ISTD would retain ownership of all virtual machines, templates, clones, and scripts/applications including application code (all versions) created for the ITAS, and all licenses purchased under the ITAS during the contractual period. MoDEE/ISTD retains the right to request (or should be able to retrieve) full copies of these virtual machines at any time
- ▶ The Winning Bidder should configure, schedule, and manage backups of all the data including but not limited to files, folders, images, system state, databases, and enterprise applications
- ▶ Perform and store data and file backups consisting of an initial full back up with daily incremental backups for files.
- ▶ For the files, perform periodic backups.
- ▶ For the databases, perform a twice weekly full database backup, with a three times daily backup of database log files
- ▶ Encryption of all backup files and data and management of encryption keys as a service that can be enabled for ITAS that require such a service.
- ▶ Winning Bidder shall not delete any data at the end of the agreement (for a maximum of 180 days beyond the expiry of the Agreement) without the express approval of MoDEE/ISTD.
- ▶ Winning Bidder to provide the RTP/RPO calculations as a part of his proposed solution.
- ▶ The Winning Bidder should ensure that the ITAS application architecture support the failover. In the event of a Primary site failover or switchover, DR site will take over the active role, and all requests will be routed through that site
- ▶ The Winning Bidder should ensure the integration of ITAS with NOC and SOC of the MoDEE/ISTD

- ▶ The Winning bidder must provide the solution hosted on the Government Private Cloud (GPC) and fully integrated with Nutanix AHV.
- ▶ The Winning bidder must provide the solution design architecture that includes the required virtual machines (VMs), databases (DBs), or any other related services (GPC and infrastructure services are listed in the below table*).
- ▶ The Winning bidder must describe each component's functionality and role in the architecture, knowing that GPC provides 3-tiers architecture for management and security purposes (Web, App, Data).
- ▶ The Winning bidder must provide the required sizing (computing specifications) that will be utilized for VMs and DBs.

The bidder is required to provide the following information in the technical proposal in relation to the required infrastructure:

- ▶ Proposed options for hosting on GPC
- ▶ Logical infrastructure architecture showing all solution components and its description
- ▶ Proposed GPC services that will fulfill project's needs and requirements
- ▶ Required computing resources to host the solution and the bidder must provide the sizing
- ▶ State compliance to submit all deliverables.

Note: If during the implementation found that the infrastructure component described in the technical proposal submitted by the winning bidder does not fulfill the requirements of the scope of this project, then the winning bidder must provide all additional needed infrastructure components and the cost of all these additional components will borne by the winning bidder

The winning bidder is required to provide the below deliverables:

- ▶ Comprehensive logical infrastructure architecture.
- ▶ backup policy
- ▶ business continuity plan
- ▶ Computing resources required to host the solution.
- ▶ All solution components and services required along with its description
- ▶ All required licenses.

4.5.3. Network Connectivity

MoDEE shall provide the connectivity between the DC and DR of the GPC. The connectivity between the landing points of the GPC in Jordan till the ISTD Headquarters (HQ) shall be the responsibility of the Winning Bidder. The Winning Bidder is expected to provide the bandwidth requirements in terms of sizing and networking equipment at the landing point and ISTD headquarters for networking solution. Winning bidder needs to ensure that all 25 branch offices of ISTD connected to HQ are able to access ITAS over a secured network.

Winning Bidder is expected to propose a networking solution with more than one existing Telecom Service Providers in Jordan. However, the final selection of a telecom service provider would be based on consultation with MoDEE/ISTD.

4.6. **Operation and Maintenance**

The Operations & Maintenance phase would be for a period of 120 months from the Go-Live date. The GoLive milestone is defined in Section 7 (Acceptance and Go-Live). A summary of O&M operations is tabulated in O&M Roles and Responsibilities Table below for reference. An indicative scope of major O&M activities is provided in subsequent sections. As for the Operations & Maintenance period, the winning bidder to ensure a legal available presence during this period. The winning bidder to either to train their own local resources or to partner with a local party.

4.6.1. **Enterprise management services**

The ITAS Technology Solution can be broadly categorized into four solution components viz. ITAS Software System, ITAS Hosting, Software Development Lifecycle & Release Management, and ITAS Operations & Maintenance.

This section provides an overview of the activities of the ITAS project relating to two broad areas: Designing to manage the entire infrastructure, application, and Operations & Maintenance of the same. As discussed above, the application is hosted in Cloud, managed by MoDEE. An indicative list of services, as well as Roles and Responsibilities are provided in the table below:

Pre-Requisites

- ▶ The Winning bidder will supply its own EMS tool and integrate the same with the EMS instance of the MoDEE.
- ▶ The rerouting of the technical queries, from the Call Center would be registered in the EMS and / or helpdesk tool of the Bidder.
- ▶ The Winning bidder shall be responsible for the integration of Call Center CRM with its proposed EMS tool
- ▶ The first level troubleshooting for technical queries shall be responsibility of the Winning bidder

Table 22: Enterprise management services

Domain	Activity – Management and Monitoring	MoDEE / Bidder
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Infrastructure	▶ Cloud Management (IaaS) ▶ System Administration ▶ Storage Administration ▶ Configuration Management ▶ Event Correlation	MoDEE
Network	▶ Landing Point connectivity at the Cloud DC ▶ Landing Point connectivity in Jordan ▶ Encryption and protecting against Distributed Denial of Service (DDoS) attacks ▶ Identity and Access Management (IAM) Roles and Groups	MoDEE

Domain	Activity – Management and Monitoring	MoDEE / Bidder
	▶ Landing Point connectivity to ISTD	Winning bidder
	▶ Network Monitoring ▶ NOC Management ▶ Integration of Bidders EMS with NOC with support of MoDEE	▶ MoDEE ▶ MoDEE ▶ Winning bidder
	▶ SOC Management ▶ Data feed into SOC ▶ Integration with SOC with Bidders EMS with support of MoDEE	▶ MoDEE ▶ MoDEE ▶ Winning bidder

Security Management		▶ Infrastructure Security Management ▶ Additional Security Requirements in consultation with MoDEE ▶ Application Security Management	▶ MoDEE ▶ Winning bidder ▶ Winning bidder
Database		Administration of database	▶Winning bidder
Backup/Replication/Restore /Archival		Application and Database	▶ Winning bidder
Demand And Management	Capacity	▶ Scalability requirements on half yearly basis ▶ Provisioning of Cloud infrastructure	▶ Winning bidder ▶ MoDEE
Incident Management		▶ Incidents related to IaaS ▶ Incidents related to application ▶ Escalation of incidents	▶ MoDEE ▶ Winning bidder ▶ Winning bidder
Service Levels		▶ Infrastructure related SLAs ▶ Application related SLAs ▶ Overall SLA management	▶ MoDEE ▶ Winning bidder ▶ Winning bidder ▶

System Software Management	License	▶ OS and other infrastructure related license management including back-up solution ▶ ITAS application related license management	▶ MoDEE ▶ Winning bidder
Domain	Activity – Management and Monitoring		MoDEE / Bidder
DC DR Management		▶ Cloud DC DR Management infrastructure layer (Active – Active or Active – Passive)	▶ MoDEE

4.6.2. Database administration

The ITAS application and corresponding databases would be hosted in the Government Private cloud managed by MoDEE. The Winning bidder is expected to finalize the requirements of Database administration during the requirement gathering phase of SDLC in consultation with MoDEE. Winning bidder's scope of work for Database Administration includes but not limited to the following:

- ▶ Winning bidder shall be responsible for monitoring database activity and performance, changing the database logical structure to embody the requirements of new and changed functionalities.
- ▶ Winning bidder shall be responsible to perform physical administrative functions such as reorganizing the database to improve performance.
- ▶ Winning bidder shall be responsible for tuning of the database, ensuring the integrity of the data and configuring the data dictionary.
- ▶ Winning bidder will follow guidelines issued by MoDEE, including access of database by system administrators and guidelines relating to security of database.
- ▶ Database administration should follow the principle of segregation of duties as specified by MoDEE to ensure no single DBA can update production tables/data singularly.
- ▶ In addition to restrictions on any direct change in data by any administrator, the Databases shall have Auditing features enabled to capture all activities of administrators.

4.6.3. SOC services

MoDEE has an existing SOC setup to monitor the security operations of the applications of Government Ministries. It is envisaged that the existing SOC would be used to monitor the ITAS application.

The Winning bidder is required to integrate the ITAS application with existing SOC for effective and real time performance monitoring. The responsibility of the Bidder includes the following:

- ▶ Define objectives, key activities, technical and physical requirements of ITAS for integration with SOC
- ▶ Conduct baseline requirement analysis of the SOC set up and provide additional enhancement augmentation requirements to MoDEE for consideration of update of SOC requirements.

- ▶ Undertake integration of tools and network architecture and prepare a process for reporting requirement for ITAS and share it with MoDEE for generation of necessary reports
- ▶ Manage events log, events incidents etc. related to ITAS, as reported in the 'VIEW' dashboard of SOC made available to Bidder by MoDEE
- ▶ Provide requirement of manpower to MoDEE for services

Note:

- ▶ The security team of MoDEE is in tendering process for security monitoring services and solution. The Bidder is required to consider the proposed solution of MoDEE and provide a security solution in the meantime, in consultation with the Security team of MoDEE.
- ▶ Also, the Winning bidder needs to budget for the Source Code review for all services in its cost.

4.6.4. NOC administration

MoDEE has an active intranet and internet network. For the ITAS project same is envisaged to be utilized for monitoring and management of Network Operations. Winning bidder is expected to understand and provide recommendation for additional requirement to augment this facility and required to integrate ITAS to this existing NOC.

- ▶ Define objectives, key activities, technical and physical requirements of ITAS for integration with NOC
- ▶ Conduct baseline requirement analysis of the NOC set up and provide additional enhancement augmentation requirements to MoDEE for consideration of updating of NOC requirements if any
- ▶ Undertake integration of tools and network architecture and prepare a process for reporting requirement for ITAS and share it with MoDEE for generation of necessary reports
- ▶ Manage events log, events incidents etc. related to ITAS, as reported in the 'VIEW' dashboard of NOC made available to Bidder by MoDEE
- ▶ Provide requirement of manpower to MoDEE for services

4.6.5. Demand and capacity management

MoDEE has an existing contractual arrangement with the Government Private Cloud, for managing the demand and capacity for Cloud hosting environment. The details of the demand management would be shared with the Winning Bidder.

However, the Winning bidder needs to undertake the following activities for managing the demand and capacity of the hosting environment for ITAS:

- ▶ As part of the technical proposal the Bidder is required to provide the capacity planning & sizing parameters for the following:
 - ▶ Server, quantity, and configuration
 - ▶ Storage requirements for both SAN and / or NAS, including the disk size
 - ▶ Network bandwidth sizing for landing point, for connectivity to ISTD
 - ▶ Virtualization requirements

- ▶ The Winning bidder shall conduct a requirement analysis and provide annual projection of the hosting requirements to MoDEE for infrastructure augmentation
- ▶ The actual deployment of the hosting requirements may be done on a quarterly, half yearly or yearly basis in consultation with MoDEE
- ▶ In case the hosting requirements are required to be scaled up or scaled down, the Bidder shall initiate the process with MoDEE accordingly

4.6.6. System software management

ITAS operations and maintenance support will include, troubleshooting, addressing availability of functionalities and performance issues and implementing change requests, license management, updated and upgrades, etc., as per requirements of ISTD

4.7. Data conversion, cleansing, migration, and transition requirements

Data Migration in the context of this RFP refers to validation and migration of data from the legacy system to the new database schema, linking and Meta tagging the documents to the relevant records in the Data Management System (DMS) and conformance to quality control requirements. Data should be migrated from the current application and media to the new application prior to 'go-live' of the application.

The Bidder should describe the process for managing the data requirements for the System, including data extraction, conversion, cleansing, migration and transition methodology and approach. The ISTD also expects the bidders to include, as part of their conversion approach, a description on how it will work with ISTD Staff to come to an agreement on various items, including but not limited to volume of data to be converted, level of detail, data objects, what data should be made available in an archived "read-only" state, etc.

The Bidder shall also highlight the proposed tools and framework to be used for seamless transition from the current ISTD eServices/Core Tax System to the proposed ITAS solution. The Bidder shall propose a solution for digitization of existing documents covering document taxonomy, metadata, classification, scanning (or conversion into soft copy), archival, storage and retrieval / readability from various structured, unstructured, and physical data sources. Since the data is also in Arabic, the bidder shall deploy requisite experts with Arabic knowledge. In order to effectively perform data assessment and cleaning, it is important that the data migration team is sufficiently familiar with existing data structures of the source registers/databases.

The Bidder should describe their data migration strategy, approach, and methodology by considering key factors including historical data, open transactions, and backlogs. The bidder should define the process which is to ensure the successful and timely conversion of data from the existing legacy system into the new proposed system. The bidders need to provide: Data Migration Design, Data Migration Execution Plan and Cutover Execution Plan. All results of the data migration test (in lower environments), and the actual migration must be preserved and accessible for future use.

The approach should include the following:

- ▶ A data conversion approach including blueprint document and plan.
- ▶ Identification of any risks associated with data conversion, cleansing and conversion.
- ▶ An approach and plan for data conversion testing. The data migration test and execution process should follow the ISTD security policy and security requirements.

- ▶ Mock data conversions and their alignment to testing activities.
- ▶ Data reconciliation reports including remediation of data to be carried out by ISTD prior migration
- ▶ Data governance and versioning.
- ▶ Data conversion mapping and transformation.
- ▶ Data conversion signoff and cutover processes.

A high-level migration project phases is shown below for consideration:

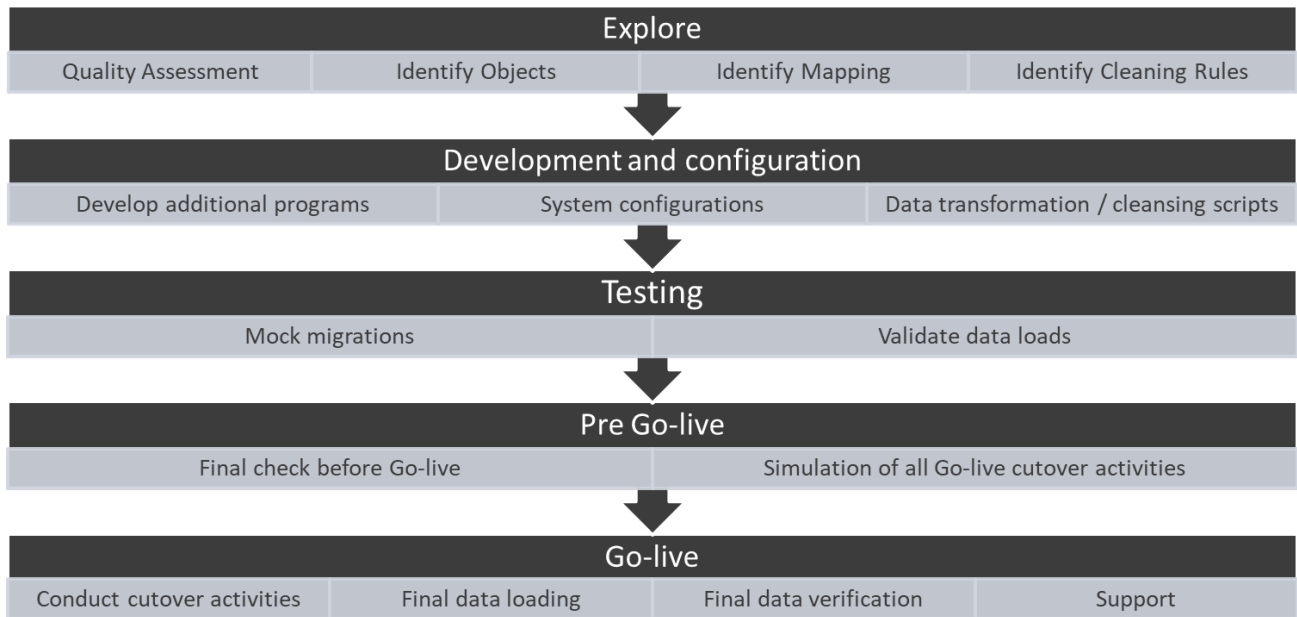


Figure 15: ITAS Migration Approach

4.8. User Experience Requirements

The ISTD expects the bidder to understand the current challenges, future requirements, and journeys of ISTD users (taxpayers, tax officers and others) and accordingly address the UX requirements in the ITAS design as per the Jordan Government Websites Standards (GWS) as explained in the ITAS Annexure H – Government Web Standard and ITAS Annexure I – Government Web Standard Checklist. Please refer ITAS Annexure B - ITAS Technical Requirements for detailed requirements.

The Jordan Government Websites Standards (GWS) make the Jordanian government websites more usable, user-centric, and universally accessible. Additionally, it brings uniformity in the quality of content and provided services. Compliance to these guidelines will ensure a high level of consistency and uniformity in the content coverage and presentation that will lead to further excellence promotion in the Jordanian Government web space, whereby all government entities are strongly encouraged to comply with these guidelines.

4.9. Organizational Change Management (OCM)

4.9.1. OCM Strategy

The bidder is expected to provide an OCM strategy to integrate OCM approaches and activities into Program activities. The OCM activities will continue throughout the entire duration of the Program, including implementation, final deployment of the System, and transition to support services. This includes confirming and validating roles and responsibilities, as well as aligning expectations of all parties through effective communications.

As part of the overall OCM strategy, the bidder will need to focus on three main OCM activities during the Program:

- Enable stakeholder engagement, including ongoing communications planning and execution. The goals are to minimize disruption, optimize adoption, and

- ▶ Enable realization of targeted benefits.
- ▶ Enable a series of iterative Organizational Change Impact Assessments (OCIAs) and Organizational Change Management Planning (OCMP) activities in parallel with each System release. The goal is to help identify process or policy changes, people or cultural impacts, and any associated role or organization changes required to enable the ISTD to adopt the System successfully and fully.
- ▶ Develop and provide appropriate training, support, and coaching to the ISTD OCM Team and Staff to facilitate the successful knowledge transfer, skills acquisition, and ultimate adoption of the System.

4.9.2. OCM Approach

The bidder needs to describe the approach to the organizational OCM process. The response should include the following:

- ▶ Methodology to provide OCM services aligned with the scope, key strategic themes/drivers, and requirements and Specifications in the RFP
- ▶ An approach and methodology for communications management, including the methods and activities needed to ensure timely and appropriate communication to Program stakeholders
- ▶ Application of industry, process, and system expertise to minimize downstream re-work and to optimize the engagement and alignment of the ISTD's key stakeholders across the System being deployed.

5. Manpower Requirements

In order to carry out the scope of ITAS project, Bidder shall be required to deploy experienced and skilled manpower as per the minimum requirements mentioned in subsequent sections of this RFP. This section outlines guidelines for staffing and provisioning of manpower for implementation of ITAS by the Bidder. The Bidder should study these guidelines carefully and adherence or deviations to any terms should be clearly highlighted in the technical bid.

Bidder is required to provide detailed information about its proposed team/staff (project manager, lead specialists, and all key team members) qualifications and experiences, included in their respective CVs. Bidder shall propose a core project management team composed of leading experts and experienced resources, who will assume overall responsibility for the implementation and management of this project.

5.1. Provision of On-Site Resources

The winning bidder shall be responsible for providing two dedicated in-house resources—a **Tax Expert** and a **Project Manager**—who will be stationed within the Ministry for the duration of the project's contract (2 years). These individuals will work under the direct supervision of the Ministry.

Key Requirements:

Placement & Supervision:

- The Tax Expert and Project Manager will be embedded within MoDEE's staff.
- They will report directly to the Ministry's designated officials

Tax Expert:

Roles & Responsibilities:

- Act as the Ministry's in-house tax advisor for ISTD related projects.
- Provide expertise on Jordanian tax laws, compliance, and regulatory requirements.
- Support project implementation by ensuring tax-related components align with local laws and best practices.
- Collaborate with ITAS project teams to analyze tax-related requirements and offer guidance.
- Liaise with relevant tax authorities and stakeholders.

Qualifications & Expertise:

- Minimum of 15+ years of experience in Jordanian tax regulations, compliance, and advisory.
- Strong knowledge in government-related tax policies.
- Strong understanding of digital tax systems is a plus.

Project Manager:

Roles & Responsibilities:

- Managing and overseeing the implementation of digital transformation projects throughout all phases of the project lifecycle, following global best practices and adhering to both PMI project management standards and agile methodology.
- Monitoring the performance of contracted bidders to ensure the execution team meets required performance standards, guaranteeing the delivery of project outputs according to requirements, work plans, and within the allocated budget.
- Preparing and reviewing work plans, periodically evaluating project performance, identifying potential risks, and developing response plans in collaboration with vendors and all relevant stakeholders.
- Developing a cash flow plan for projects, aligning financial flows with performance progress according to objectives and work plans.
- Monitoring, controlling, documenting, and managing changes that occur during project execution in accordance with the ministry's established procedures and guidelines.
- Managing and organizing relationships and communication with internal and external project partners to ensure effective collaboration and achieve the desired project outcomes.
- Preparing periodic progress reports on project execution for all relevant stakeholders, in coordination with the implementing company.
- Ensuring and following up on the implementation of all acceptance tests related to project deliverables, including quality and information security standards.
- Overseeing the activation and launch of e-services within the relevant organization, as per the project scope and documentation.
- Contributing to the review of digital transformation project tender documents in collaboration with other concerned departments within the Digital Transformation and Enablement Administration to

ensure compliance with work procedures.

- Performing any other tasks as required by the nature of the work.

Qualifications & Expertise:

- 7+ years in ICT project management, preferably in digital transformation and ERP projects.
- Project planning, scope definition, best practices, and implementation management.
- Ability to manage multiple projects simultaneously under pressure.
- Experience managing diverse teams, particularly in government settings.
- Strong critical thinking skills for complex issue resolution.
- Fluent in Arabic and English (verbal & written).
- Strong presentation, facilitation, and motivation abilities.
- Detail-Oriented: High attention to accuracy and precision.

Costs & Responsibilities:

- The bidder shall cover all costs related to salaries, benefits, and obligations of the in-house resources.

MoDEE will provide workspace, access to necessary tools, and facilitate their integration into the team

5.2. Deployment of Resources

Key resources of the bidder would need to be on-site to work closely with ISTD to achieve the projects objectives. The key personnel comprise of program manager, project manager, UI/UX Expert, solutions architect, integrations architect, infrastructure & networks expert, and tax domain expert. These key resources should work onsite for the duration of the project. The onsite requirement for O&M phase to be mutually agreed with ISTD at all times. Based on their proposed approach and in line with the key resource requirements, bidder is expected to provide a detailed manpower deployment plan.

In the response to this RFP, Bidder shall propose only one candidate for each key position. The Project Team shall at a minimum include expertise and experience in the following disciplines:

Table 23: Key Resources of Bidder

Sr. No.	Role	Experience	Arabic Speaker (Yes/Preferred)	Certification (Preferred)
1	Engagement Director	15-18 years	Preferred	PMP/ Prince 2 (Mandatory)
2	Program Manager	12-15 years	Yes	PgMP/ PMP/ Prince 2 or equivalent (Preferred), Agile certified or Scrum Master or equivalent (Mandatory)

3	Project Manager	10 – 12 years	Yes	PMP/ Prince 2/ Agile certified or equivalent
4	Expert - Infrastructure & Networks	10-12 years	Preferred	AWS, Azure, GCP Cloud certified

5	Team lead – Testing	10 – 12 years	Yes	International Software Testing Qualifications Board (ISTQB) certifications
6	Domain expert/ Business Analyst (Tax Specialist)	10 – 12 years	Yes	TADAT certified
7	Solutions Architect	10 – 12 years	Preferred	TOGAF, COBIT
8	Database Architect	8-10 years	Preferred	AWS, Azure, GCP Cloud certified in database specialty
9	BI and Analytics Expert	8-10 years	Preferred	AWS, Azure, GCP Cloud certified in BI/Analytics specialty
10	Module wise Team Lead	5+ years	Preferred	
11	Team Lead– Business Intelligence and Analytics	5+ years	Preferred	
12	Operations Manager	8-10 years	Preferred	
13	Security Architect	8-10 years	Preferred	CISSP, CISM, CISA certified
14	Training Expert	8-10 years	Yes	IBCT Certification for Trainers
Sr. No.	Role	Experience	Arabic Speaker (Yes/Preferred)	Certification (Preferred)
15	Quality Analyst	8-10 years	Preferred	CSQA

16	UI/UX Expert	8-10 years	Yes		
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17	Systems Integration Architect	8-10 years	Preferred	TOGAF, SOA/Middleware/API certifications
18	Tax Technical bilingual Experts (Arabic & English)	5+ years	Yes	
19	Change Management Specialist System	8-10 years	Preferred	

Bidder shall ensure the availability of key project staff, especially Team Leads for each of the disciplines listed mentioned above and must be certified as well as personally signed by the team member.

Key resources of the bidder need to be onsite during different phases of the project to ensure the smooth and success implementation of the project. Bidder shall provide a detailed resource plan and deployment plan for all project phases.

Additionally, depending on the project phase MoDEE/ISTD might ask for specific resource(s) to be available onsite. Failing to comply with onsite presence requirement by the bidder would be subject to penalty as per the man month rate quoted by the bidder. Bidder to confirm their resource deployment plan accordingly.

5.3. Guidelines for staffing and provisioning of manpower

- ▶ While a dedicated core team shall be stationed at ISTD's premises, the actual development will happen in offshore location / development environment set up by the Bidder. Considering the implementation plan of ITAS it is expected that the Bidder will suitably staff the ITAS development and maintenance team.
- ▶ The Winning bidder shall provide a detailed staffing schedule in their Technical Proposal as per the format provided as part of this RFP.
- ▶ The staffing schedule should also include an Organization Chart showing the proposed governance structure to be established by the Bidder for execution of the scope of work. The organization chart should clearly bring out variations to the Organization structure if any envisaged by the Bidder for various phases of the project.
- ▶ Detailed CVs should be provided for key profiles including the in-house resources that will be subject to evaluation during the Bid Process Management for selection of Bidder. CVs should be as per the CV format given in this RFP. Area of expertise, role and tasks assigned should be clearly identified for each of the key profiles. ISTD might interact with the proposed resources and this interaction shall be duly considered during the technical evaluation.
- ▶ Winning bidder shall ensure that at least a Director level person is nominated to be part of the Project Steering Committee and a senior person equivalent to the rank of National / Country Manager is

nominated to be part of the Project Working group. Keeping in view the importance of the project the National / Country Manager of Bidder shall be regarded as the project owner for ITAS.

- ▶ Key project staff shall be full-time employees of the Winning bidder firm.
- ▶ The Staffing Schedule should contain the schedule of deployment of the key resources. It should also clearly highlight onsite and offsite deployment of each profile.
- ▶ In-house resources may be required to undergo interviews conducted by MoDEE to assess their qualifications following the selection of the winning bidder.

5.4. Replacement of Personnel

- ▶ The Winning bidder should avoid any change in the organization structure and proposed manpower as part of the scope of services.
- ▶ If the same is however unavoidable due to exigencies, Winning bidder shall promptly inform ISTD in writing for due approval. Further Winning bidder shall ensure that they provide a suitable resource of equal or better qualification and experience as a replacement. All such cases would be reviewed by ISTD and may interact with resource before approving replacement of any core team member.
- ▶ In case of such replacement of any key resource, the Winning bidder should ensure proper knowledge transfer from the outgoing resource to the incoming resource and adequate hand-holding period / training for the incoming resource to maintain the continued level of service.

5.5. Logistics and Removal of Personnel

- ▶ The Bidder shall be responsible for the deployment, transportation, accommodation, and other requirements of all its resources deployed for the execution of the scope of work and provision for all costs/charges in connection thereof.
- ▶ During the engagement, MoDEE/ISTD may request removal of any team member or any person(s) deployed by the Winning bidder about professional incompetence or negligence of work or for being deployed for work for which he is not suited. Bidder shall take up all such requests and ensure replacement of resources as required, with a competent substitute at no additional cost to MoDEE/ISTD
- ▶ In case the Winning bidder wishes to remove / replace any key resource for reasons other than exigencies, Winning bidder should submit this request in writing describing the reasons and replacement options for consideration of MoDEE/ISTD. MoDEE/ISTD at its own discretion may choose to either accept or reject such requests depending upon the merit of the case.

5.6. Escalation Matrix

- ▶ As part of the technical proposal, the Winning bidder shall provide a detailed Escalation Matrix mapping back to the Bidder's proposed organizational structure for the entire implementation phases.
- ▶ The Escalation Matrix should address key requirements stated in the Service Level Agreements for various service delivery activities and cover all major service delivery activities.
- ▶ The triggers for escalation should be clearly identified and stated for each category of service in the Technical Proposal.
- ▶ The Winning bidder will also prepare the Escalation Matrix for support phase of ITAS project in consultation with MoDEE/ISTD.

6. Acceptance Criteria and Go Live

The Winning bidder shall demonstrate the testing criteria outlined in this document prior to Preliminary Acceptance / Go-Live for each Phase as well as during project operations & management phase. In case required, parameters might be revised.

Following activities needs to be completed by the Winning bidder:

- ▶ Prepare and agree with ISTD a detailed Go Live Plan in line with the implementation schedule as mentioned in the Section 8
- ▶ Complete the Data Migration activities
- ▶ Testing as per standard SDLC, including but not limited to
- ▶ System Integration Testing of the ITAS as whole along with 3rd party systems
- ▶ Performance and load testing
- ▶ Security testing (including penetration and vulnerability testing)
- ▶ Assistance in User Acceptance Testing of the ITAS
- ▶ Creation of all the environments mentioned in this document
- ▶ Completion of the total sessions of training agreed under the contract
- ▶ Get Clearance for the Beta Testing

Note: On issuance of final Preliminary Acceptance certificate by ISTD/MoDEE at the end of implementation, the Operations & Management phase of the ITAS project will commence.

7. Project Management and Governance

7.1. Project Management Office (PMO)

The Ministry of Digital Economy & Entrepreneurship is following the PMI standards for managing projects and as per the PMI best practices.

In order to provide project management services, the Winning bidder is required to setup a Project Management Office to ensure consistency and standardization of process, implementation, and communication across all projects of ITAS. The PMO should focus on increasing interaction of people, process, and tools for effective collaboration, communication, accountability, and control. The Bidder should provide all activities and processes needed for the proper functioning of the project implementation.

The PMO will appoint a designated full time Project Manager to oversee the project execution together with project teams to track and execute all designated tasks and activities. The Bidder should develop a Project Plan, which includes project objectives, deliverables, role & responsibilities, communication protocols, document control methodology, cost management, schedule management, quality management plan and define the required success criteria. The PMO team shall, as required, constitute other relevant stakeholders, individuals from project working groups and other representatives by invitation. The key roles and objectives of the PMO are, but not limited to:

- ▶ Develop a detailed Project Plan, including project objectives, Deliverables Traceability Matrix (DTM), project schedule, roles & responsibilities, communication protocols, document control methodology and quality management. The plan should also ensure that all project objectives are achieved within the overall project schedule.
- ▶ Develop a project implementation strategy based on the requirements of MoDEE/ISTD and consider a phased approach of implementation if required.
- ▶ Organize weekly formal meetings to track project progress, address issues and concerns, open and escalated issues, highlight project risks, discuss required mitigation plan, assess timelines and identify any potential delay in deliverables, and discuss proposed changes, if any. The agenda of the meeting shall not be limited to the listed points.
- ▶ Review and verify the progress of the project team's tasks & deliverables in accordance with the project schedule and DTM.
- ▶ Conduct frequent meetings with the project team and track progress.
- ▶ Record and distribute Meeting Minutes detailing the key decisions taken, actions items and points discussed.
- ▶ Conduct periodic progress meetings with all stakeholders on a monthly basis to cover project progress in a comprehensive manner.
- ▶ Introduce processes for reporting project progress, issues, delays in deliverable submission, project risks and issue management.'
- ▶ Prepare performance and SLA compliance reports.
- ▶ Maintain track of development status, testing results, deployment, and operation status at a project level.

- ▶ Manage technical and business scope change across ITAS project and execute proper change management.
- ▶ Resolve differences in approach, positions, opinions, and interpretations that can occur in the project.
- ▶ Introduce PMO Tools to support and effectively manage efforts.

7.2. Project plan

Describe the proposed approach to plan, manage, and complete the activities and Deliverables based on the requirements in the RFP. The Bidder must propose a consistent approach for planning and tracking of projects. The response must include, but is not limited to:

- ▶ Project Plan outlining the Bidder's approach for managing the project and key Deliverables
- ▶ Project management methodology to be followed for the implementation
- ▶ Deliverables (Arabic and/or English) required for successful completion of the project and the criteria for approval of deliverable.
- ▶ Proposed timeframe and release schedule based on the requirements and specifications in the RFP.
- ▶ Approach for risk analysis, risk prioritization, risk mitigation, and risk tracking, including tools and techniques that should be used for risk management and escalation.
- ▶ Approach and methodology for issue management to ensure that issues are identified, evaluated, assigned for resolution, and resolved. Include tools and techniques that should be used for issue management and escalation
- ▶ Approach and methodology for change control to address changes to the scope, cost, and schedule, including tools for tracking and approval of change requests.
- ▶ The project plan must ensure that all project objectives are attained through proper prioritization taking into consideration the requirements in the RFP.

7.3. Project status monitoring and reporting

The bidder must establish a systematic process for tracking and reporting project progress including timelines, delays and issues whereby ensuring that deliverables are met within timelines. Propose efficient tools to monitor the status of activities and provide a consolidated view of progress across the project. Status reports should be circulated within the project team and presented in the progress meetings.

7.4. Risk and issue management

Implement a project-wide risk planning and management process and ensure the mechanism for dealing with issues is efficient and robust. Propose quality measures and establish a dedicated quality assurance function that helps identify risks and resolve key issues. Monitor and track risks/issues associated with the project for timely intervention and mitigation.

Define an approach to analyze, prioritize, mitigate, track and control risk. Conduct risk assessments to identify security threats and risks to the system. Define a methodology for issue management to ensure that issues are identified, monitored, analyzed, and resolved. Propose tools and techniques for issue and risk management and escalation.

7.5. Change control management

The bidder is expected to formulate a change management strategy and work with the project team to integrate change management approaches into project activities. This includes aligning expectations of all stakeholders through effective communication, planning, and execution. The goals shall be to minimize disruption, optimize adoption, and enable realization of targeted outcome. Define and manage scope. Ensure that the impact of proposed changes is understood, and the agreed changes are tracked and managed. Maintain a provision to account for variables (like scope creep) impacting budget & time. Propose a methodology for change control to address changes and suggest tools for tracking and approval of change requests.

7.6. SLA management

The Winning bidder is required to prepare the service levels required for operations of the system. The bidder is responsible for the deployment of an SLA management tool and the measurement of the SLAs on a periodic basis.

7.7. Transition and exit management

The Winning bidder should submit a detailed transition and exit management plan for ITAS which shall be discussed and approved by MoDEE/ISTD. It is required to develop and execute training plan and knowledge transfer for the identified team for transition. The transition plan shall cover major components of the installed solution, including, but not limited to system administration, system monitoring, data management, reports, and dashboards. The handover shall also include transfer of knowledge related to system installation, operation, backup, restoration, troubleshooting and failure recovery protocols. All risks during the transition stage should be documented and mitigation measures should be defined to ensure a smooth transition without disruption.

The transition and exit management plan require to be updated on a periodic basis and the updates needs to be reviewed and approved by MoDEE/ISTD. The bidder should ensure that necessary knowledge transfer and transition support are provided at the time of transition. The deliverables for transition are, but not limited to:

- ▶ Updated transition plan approved by MoDEE/ISTD.
- ▶ System documentation and support related documents.
- ▶ List of inventories of all assets created for the project.
- ▶ Detailed explanation and demos for the system.
- ▶ Hand-over of the entire software including source code, program files, configuration files, setup files, project documentation, credentials etc.

The bidder shall be released from the project once the transition is completed successfully, and the required criteria defined for successful transition are met.

7.8. Nearshore/Offshore Delivery Centre visits

MoDEE/ISTD may visit the shortlisted bidders' delivery centers to understand their delivery capabilities, IT infrastructure including cyber security compliance, work culture, adherence to local laws etc. Similar visits may happen during the project implementation as well at important milestones. Bidder is expected to facilitate such visits to their premises that provide confidence to MoDEE/ISTD in achieving program objectives.

8. Implementation Schedule

8.1. Project Timeline, Deliverables and Milestones

8.1.1. Project Timeline and Deliverables

The detailed implementation schedule with milestones and activities related to the ITAS project is presented below. The Bidder must ensure that all timelines are adhered to for a successful rollout of the project. Please refer to Annex M.

Table 24: Project Timeline

Activity	Deliverables	Expected Timelines (In weeks)	Completion
Project Inception	Kick-off presentation and/or Duly signed agreement	T	
	Submission of project charter. It should cover the following: ▶ Setting up of Program/Project, requirement and change management and collaboration tools ▶ Study of scope of work & functional coverage ▶ Governance Structure for Project Implementation ▶ Project implementation approach ▶ Detailed Project Plan with work breakdown structure (including resource deployment plan) ▶ Delivery schedule ▶ Key milestones ▶ Resource deployment plan ▶ Change & communication management plan ▶ Change control procedure ▶ Exit management plan ▶ Draft BRD and SRS template	T + 2	
	Deployment of key personnel & In-house Resources	T + 2	

Requirements Gathering	Gap Analysis Report which covers the following: ▶ Workshops for bidder's tailored process maps walkthrough to ISTD (based on bidder's solution, ITAS RFP requirements and Jordan Tax policy and legislations) ▶ Workshops for bidder solution's pre-configured User Personas and associated journeys maps, Clickable Prototypes for Taxpayer and Tax officer portals and mobile apps, Portals and Mobile apps User Interface Usability Report (based on Jordan's web usability guideline and policy) ▶ Finalization of bidder's tailored process maps, User Personas and associated journeys maps, best practices, Clickable Prototypes applicable for ITAS as it is, and which needs to be localized/customized as per Jordan and ISTD requirements/policies ▶ Processes for which extensions (localization/customization are required)	T + 6
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Activity	Deliverables	Expected Completion Timelines (In weeks)
	▶ Gap Analysis report sign-off from ISTD	

Architecture and Design	▶ Detail integration requirements with MIS / Reports / dashboards of inter & intra Ministries (API details, data fields to be shared, data fields required from other ministry systems, process flows, flow charts, design diagrams, etc.) shared with all stakeholders ▶ Clickable Prototypes of all persona screens including taxpayer, tax officer, government agency users ▶ SRS and Prototypes sign-off from ISTD	
	ITAS Dashboard and analytics report ▶ List of MIS / Reports with recommended visualizations ▶ List of external data point integrations with detailed integration process flows and documentation ▶ Data Analysis tools, techniques, and usability documentation ▶ Data Analysis output / parameters / MIS / Reports ▶ User access and dashboard MIS mapping matrix ▶ ITAS Dashboard and analytics report sign off by MoDEE/ISTD	T + 24
	System Design & Configuration report should cover the following: ▶ System Architecture Document ▶ System Configuration and module wise configuration needs as per the design envisaged ▶ Legacy and Third-party System Integration / interface Report and integration of same with the envisaged solutions ▶ Customization Development Plan and Design / development plan of components of functionalities ▶ High Level Software Design document including Software Architecture design, Logical and Physical Database Design ▶ Low Level Software Design document including Programming Logic, Workflows ▶ Deployment Architecture Document ▶ Finalization of KPIs / KRAs for dashboards ▶ Wireframe/Prototype design of dashboards	T + 28

Activity	Deliverables	Expected Completion Timelines (In weeks)
Implementation	▶ System architecture, Design & Configuration sign off by ISTD ▶ Development and testing of ITAS (divided into max. 3 deliveries – each release containing multiple sprints/iterations, show/tell demo sessions to ISTD, SIT, UAT. Bidder to detail these 3 deliveries in their project plan) ▶ Software Deployment report (for each release) should cover the following: ▶ Complete Source Code with documentation ▶ Dashboard and analytical tool deployment with data validation report ▶ Test Plans and Test cases (including Unit Test Plan, System / Integration Test Plan, User Acceptance Test Plan, Security Test Plan, Load Test Plan) ▶ Software Testing Documentation (including details of defects / bugs / errors and their resolution) ▶ Unit Test Cases, System Test Results / Cases ▶ System Test report ▶ Integration Test with 3rd parties Report ▶ System Integration Test (SIT)/End to End Test Report ▶ Requirement Traceability Matrix (RTM) ▶ User Acceptance Test strategy and execution plan ▶ User Acceptance Test Preparation (Multiple UAT preparation based on releases) - User Acceptance Test Cases, Test Data setup, User Acceptance Test Scripts	T + 68
	UAT support activities by bidder including defect resolution, supporting ISTD staff in testing (Iterative UAT based on different releases and a final consolidated UAT) for all the 3 deliveries in the implementation phase. A consolidated UAT must be planned after the successful UAT of each of the three deliveries.	T + 88

	Deployment of production licenses and completion of environment readiness	T + 88
	Beta Testing completion	T + 96

Activity	Deliverables	Expected Completion Timelines (In weeks)
	Benchmarking report, performance testing and security tests (including ISO 27001 certification)	T + 96
	Production deployment	T + 103
	Migration of existing data from e-services/Core Tax System to production	T + 103

	Go-live report should cover the following: ▶ UAT sign-off ▶ Complete updated Source Code and updated Deployment script with documentation ▶ Data migration sign off from ISTD ▶ Deployment sign-off from ISTD ▶ User Manuals and System Manuals sign off from ISTD ▶ Sign-off from ISTD on Dashboard and Data Analytics requirements ▶ ISTD approved Security Testing, Load Testing, Unit Testing and System Acceptance report ▶ DB entity relationship diagram ▶ Pending Issues in the system, dependencies ▶ Updated System Design documents, specifications for every change request ▶ Updated user Manuals, administration manuals, training manuals ▶ Go-Live Certificate from ISTD indicating readiness for roll-out with trainings ▶ Certification of successful completion of system stabilization from ISTD (Post go live)	T + 104
Content Finalization		
Content	▶ Content Finalization report (screen labels, tool tips, error/information/warning, user guidance content, correspondences, newsletters – emails, SMS, letters, notices, certificates, and any other relevant content identified during the project, relevant templates – English and Arabic)	T + 48

Activity	Deliverables	Expected Completion Timelines (In weeks)
Training and Capacity Building	▶ Content finalization report sign off by MoDEE/ISTD	

Training	Change Management & Training plan and strategy should cover the following:	
	▶ Detailed Training plan, ▶ Communication plan and strategy (for all stakeholders including taxpayers)	T + 48
	▶ Detailed Training curriculums on functionalities of the application and the underlying technologies to stakeholders	T + 52
	▶ Training material preparation (CBTs, Presentations, User manual documents, videos with voice over, Help modules, etc.)	T + 92
	Training to Government stakeholders (Arabic and English)	T + 102
	Change Management & Training completion for Go-live report should cover the following:	
	▶ Training session-wise completion reports ▶ Submission of Final Training Documents ▶ Certification from ISTD officials confirming successful completion of Change Management & Trainings	T + 103
Data Migration		
Documentation	Data migration report should cover the following:	
	▶ Data migration assessment ▶ Migration & transitioning approach ▶ Detailed data migration plan ▶ Scripts required for importing data that must be migrated ▶ Data back-up and archival process document	T + 24
	Mapping document for migration	T + 32

Activity	Deliverables	Expected Completion Timelines (In weeks)
Migration	Data validation and cleaning	T + 40
	Deployment of required tools and Data Migration (over iterative releases)	T + 72
	Data migration completion report (multiple Production simulation runs) should cover the following: ►Details of actual data that has been migrated ► Detailed methodology used for data migration with flow charts, size of data migrated, data validation reports. ►Certificate from ISTD officials confirming successful completion of data migration	T+ 96
	Data migration completion report (Production) should cover the following: ► Details of actual data that has been migrated ► Detailed methodology used for data migration with flow charts, size of data migrated, data validation reports. ► Certificate from ISTD officials confirming successful completion of data migration	T + 103
IT Helpdesk		
Set up	Setting up of required infrastructure for IT Helpdesk	T + 80
	Integration with the main solution and testing	T + 96
Team deployment	Deployment and Training to the helpdesk team	T + 100
IT helpdesk Go-Live	Go-Live of IT helpdesk	T + 104

Operations and Maintenance		
Processes as per ITSM standards O&M ITAS	for of Setup Processes, procedures, Policy framework as per ITSM standards for the ITAS O&M Set up	T + 104
Activity	Deliverables	Expected Timelines (In weeks)
Quarterly reports (within 2 weeks after end of month)	SLA Compliance Reports should cover the following: ▶ Performance Monitoring reports for system ▶ SLA Compliance Reports ▶ Count of SMS and Emails sent ▶ Training session-wise completion reports ▶ Patches / Upgrades of all components ▶ Incremental updates to solution ▶ Audit / Standard Compliance Reports	Ongoing
	Project Status Report ▶ On-Going Project Updates	Ongoing
	Change Control Note and Report ▶ Change Requests Managed	Ongoing
	Issue/Incident reports ▶ Issue / Problem / Bugs / Defect Tracker	Ongoing

Note:

1. T –Contract Award (Issuance of the Letter of Award) date.
2. For all the deliverables submitted by the Winning Bidder, the ISTD will approve/reject/provide feedback to the Winning Bidder within 15 working days after submission of a deliverable.
3. The Winning Bidder needs to consider and deploy independent teams for each track and phases as detailed above and provide the details of the same in the proposal and in the team deployment plan.

9. Annexures

9.1. ITAS Annexure A - Functional Requirements for ITAS

Enclosed with this document.

9.2. ITAS Annexure B - Technical Requirements for ITAS

Enclosed with this document.

9.3. ITAS Annexure C – ISTD Interfacing Systems

Enclosed with this document.

9.4. ITAS Annexure D – Data Points for ITAS

Enclosed with this document.

9.5. ITAS Annexure E – ITAS Technology Stack

Enclosed with this document.

9.6. ITAS Annexure F – IDM

Enclosed with this document.

9.7. ITAS Annexure G – Government Service Bus

Enclosed with this document.

9.8. ITAS Annexure H – Government Web Standard

Annex H- Government Web Standard [GWD-Designkit update6](#)

9.9. ITAS Annexure I – Government Web Standard Checklist

Enclosed with this document.

9.10. ITAS Annexure J – MODEE Customer Journey Standards Compliance Report 2020

Enclosed with this document.

9.11. ITAS Annexure K - Performance Test Checklist

Enclosed with this document.

9.12. ITAS Annexure L - e-service Performance KPI's

The below stated MODEE KPIs values to be achieved over internet for non-cached pages, cached numbers should be much less, also many on the numbers depends on gateway bandwidth at MODEE. The below sections act as the accepted thresholds for all MODEE developed websites and vendor should assure they are in comply with these guidelines before requesting Performance and stress test from MODEE.

Performance Metrics for E-services Criteria / Element	Description	Value / Measurement
Time to First Byte	Time elapsed for the first byte of a website to make it to the visitor's browser	Less than 3 sec
Time to Last Byte	Time elapsed when every byte of a website has made it to the visitor's browser	Less than 6-9 sec
Time to Connect	Time elapsed from initial request to when the connection between the visitor's browser and an origin server is established	Less than 2 sec
Page Load Time (for both E-services and Informational Wesbites)	Page load time is the average amount of time it takes for a page to appear on your client's screen. To measure page load time, you should be testing website speed using available tools.	Largest Concertful Paint (LCP): should occur within 3 seconds of when the page first starts loading. First Input Delay (FID): pages should have a FID of 200 milliseconds or less. Cumulative Layout Shift (CLS): pages should maintain a CLS of 0.1. or less.
Time to Start Render	Time elapsed when the first visible element appears on the blank page	Less than 6 sec
Throughput:	The quantity of useful work	[15 request/second]

Content Complexity Metrics for E-services

Criteria / Element	Description	Value / Measurement
Capacity	The capability of the newer system to handle a number of simultaneous requests from the network for the application and the volume of data that it can handle from each of the users (Internal users	

	through the LAN as well as external users through the internet/ dedicated WAN). In addition to the H/W capacity such as processing capability of all servers including DB, Apps. [CPU Utilization: 80%, Memory Utilization: 80%.	
Weight Basis		
Page Weight	Total weight of assets including requests, domains, HTML, JavaScript, CSS, images, media and others	Not exceeding 2 – 4.5 MB
JS Weight	Weight of Java Scripts	Not exceeding 1600 KB
CSS Weight	Weight of Cascading Style Sheets	Not exceeding 1800 KB
Image Weight	Weight of site images	Not exceeding 1.75- 2.5 MB
Count Basis		
Asset Count	Total number of assets including requests, domains, HTML, JavaScript, CSS, images, media and others	Not more than 100
Image Count	Images count in a site	Less than 50

- For Informative websites that need to have higher asset resources CDN network should be used.
- For any web services that used Video asset CDN network should be used.

Site Availability and Access Metrics:

Criteria / Element	Description	Value / Measurement
Uptime (Availability) in stress test	System availability uptime vs. system downtime within Stress test	99 %
Active Users	The number of concurrent users that the website should be handled during the load test, with lowest error rate ratio.	The number of concurrent users will be determined during the project initiation phase. If not mentioned, the default value will be 200 users. Error rate less than 0.9%

9.13. ITAS Annexure M - ITS Program Plan

Enclosed with this document.

9.14. ITAS Annexure N – Additional Requirements

Enclosed with this document